
R-oyalty Project 4

Elizaveta Udalova

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PROJECT 1

Topic

Public Perceptions and Responses to COVID-19 in Netherlands

Research Question

To what extent are trust in government institutions and satisfaction with the healthcare system associated with the likelihood of COVID-19 vaccination in Netherlands?

Relevance

Examining the relationship between trust in government institutions, satisfaction with the healthcare system, and the likelihood of COVID-19 vaccination is particularly relevant in the Dutch context, where public perceptions played a crucial role in shaping vaccine uptake. The Netherlands offers a compelling context for this research due to its “intelligent lockdown” approach, which emphasized individual responsibility and trust in citizens (van Dullemen & de Bruijn, 2021). A survey study published in JAMA Network Open found that trust in the government and misinformation about COVID-19 vaccines were key determinants of vaccine hesitancy in the Netherlands, indicating that skepticism toward official information could reduce willingness to get vaccinated (Yousuf et al., 2021). Investigating these associations within the Dutch context allows for a deeper understanding of how institutional and healthcare-related perceptions shape health-related behaviors, offering important implications for public health strategies in future.

Loading required libraries

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(kableExtra)
library(psych)
library(car)
library(stringr)
library(coin)
library(lsr)
library(effsize)
library(effectsize)
library(rstatix)
library(sjPlot)
library(cowplot)
library(ggcorrplot)
library(forcats)
library(nortest)
library(MASS)
library(lmtest)
library(moments)
library(interactions)
```

Description of the variables

```
covid_ned <- read.csv('/Users/ludamuraveva/Desktop/DA/data/ESS11-subset.csv', sep = ',')

covid_ned_selected <- dplyr::select(covid_ned, respc19a, symtc19, symtnc19, vacc19, psppsgva, stfhlth, stfgov, health)

variable_names <- c("respc19a", "symtc19", "symtnc19", "vacc19", "psppsgva", "stfhlth", "stfgov", "health")
meanings <- c("Respondent had coronavirus", "Respondent had symptoms lasting 3 months or longer", "Respondent still has symptoms", "Respondent received at least one dose of a vaccine against coronavirus", "State of health services in country nowadays", "Political system allows people to have a say in what government does", "How satisfied with the national government", "Subjective general health")
types <- c("Nominal", "Nominal", "Nominal", "Nominal", "Nominal", "Quasi-interval", "Quasi-interval", "Quasi-interval")

table1 <- data.frame(
  Variable = variable_names,
  Meaning = meanings,
  Type = types,
  stringsAsFactors = FALSE
)

kable(table1) %>%
  kable_styling(bootstrap_options=c("bordered", "responsive", "striped"), full_width = FALSE)
```

Variable	Meaning	Type
respc19a	Respondent had coronavirus	Nominal
symtc19	Respondent had symptoms lasting 3 months or longer	Nominal
symtnc19	Respondent still has symptoms	Nominal
vacc19	Respondent received at least one dose of a vaccine against coronavirus	Nominal
psppsgva	State of health services in country nowadays	Nominal
stfhlth	Political system allows people to have a say in what government does	Quasi-interval
stfgov	How satisfied with the national government	Quasi-interval
health	Subjective general health	Quasi-interval

Filtrating the data

```
covid_ned_selected <- covid_ned_selected %>%
  mutate(
    stfhlth = as.numeric(as.character(stfhlth)),
    health = as.numeric(as.character(health)),
    stfgov = as.numeric(as.character(stfgov)),
    respc19a = ifelse(respc19a %in% c(7, 8, 9), NA, respc19a),
    respc19a = factor(case_when(
      respc19a == 1 ~ "Yes, tested positive",
      respc19a == 2 ~ "Yes, not tested",
      respc19a == 3 ~ "No",
      TRUE ~ NA_character_
    ), levels = c("Yes, tested positive", "Yes, not tested", "No")),
    vacc19 = ifelse(vacc19 %in% c(7, 8, 9), NA, vacc19),
    vacc19 = factor(case_when(
      vacc19 == 1 ~ "Yes",
      vacc19 == 2 ~ "No",
      TRUE ~ NA_character_
    ), levels = c("Yes", "No")),
    psppsgva = ifelse(psppsgva %in% c(7, 8, 9), NA, psppsgva),
    psppsgva = factor(psppsgva,
      levels = 1:5,
      labels = c("Not at all", "Very little", "Some", "A lot", "A great deal"),
      ordered = FALSE),
    stfhlth = ifelse(stfhlth %in% c(77, 88, 99), NA, stfhlth),
    health = ifelse(health %in% c(7, 8, 9), NA, health),
    stfgov = ifelse(stfgov %in% c(77, 88, 99), NA, stfgov),
    symtc19 = ifelse(symtc19 %in% c(6, 7, 8, 9), NA, symtc19),
    symtc19 = factor(case_when(
      symtc19 == 1 ~ "Yes",
      symtc19 == 2 ~ "No",
      TRUE ~ NA_character_
    ), levels = c("Yes", "No")),
    symtnc19 = ifelse(symtnc19 %in% c(6, 7, 8, 9), NA, symtnc19),
    symtnc19 = factor(case_when(
      symtnc19 == 1 ~ "Yes",
      symtnc19 == 2 ~ "No",
      TRUE ~ NA_character_
    ), levels = c("Yes", "No"))
  ) %>%
  drop_na(respc19a, vacc19, psppsgva, stfhlth, health, stfgov, symtc19, symtnc19)
```

Checking variables' types

```
str(covid_ned_selected)
```

```
## 'data.frame': 298 obs. of 8 variables:
## $ respc19a: Factor w/ 3 levels "Yes, tested positive",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ symtc19 : Factor w/ 2 levels "Yes","No": 1 1 1 1 1 1 1 1 1 1 ...
## $ symtnc19: Factor w/ 2 levels "Yes","No": 1 1 2 2 2 2 2 1 2 1 2 ...
## $ vacc19 : Factor w/ 2 levels "Yes","No": 1 2 1 1 1 1 1 1 1 1 ...
## $ psppsgva: Factor w/ 5 levels "Not at all","Very little",...: 1 2 3 2 3 2 3 3 2 2 ...
## $ stfhlth : num 0 7 1 2 7 5 7 7 7 8 ...
## $ stfgov : num 0 5 4 2 0 5 7 5 5 4 ...
## $ health : num 2 2 2 2 2 2 3 2 3 3 ...
```

Descriptive statistics

```
explanations <- c(
  "Mode indicates that most of respondents had coronavirus and tested positive.",
  "Mode \"Yes\" means that most of respondents had symptoms of coronavirus lasting 3 months or longer.",
  "Mode \"No\" means most of the respondents do not still have symptoms of coronavirus.",
  "Mode shows that most of the respondents received at least one dose of a vaccine.",
  "The mode is 3, so most respondents chose the answer right in the middle - the political system in [country] allows people like them to have some say in what the government does.",
  "The most frequent estimation of health services in country is 8 (10 is extremely good), but a significant amount of respondents estimate health services lower because the mean answer is 6 and median is 7.",
  "Mode, mean and median are very close to each other, which means that data about satisfaction in government is quite normally distributed and most people are medium satisfied.",
  "Subjective health statistics are similar; health is good for most of the respondents.")

calculate_mode <- function(x) {
  ux <- unique(x)
  tab <- tabulate(match(x, ux))
  mode_val <- ux[which.max(tab)]
  return(mode_val)
}

modes <- sapply(covid_ned_selected, function(x) {
  mode_val <- calculate_mode(x)
  as.character(mode_val)
})

means <- sapply(covid_ned_selected, function(x) {
  if(is.numeric(x)) round(mean(x, na.rm = TRUE), 6) else NA
})

medians <- sapply(covid_ned_selected, function(x) {
  if(is.numeric(x)) median(x, na.rm = TRUE) else NA
})

ranges <- sapply(covid_ned_selected, function(x) {
  if(is.numeric(x)) diff(range(x, na.rm = TRUE)) else NA
})

variances <- sapply(covid_ned_selected, function(x) {
  if(is.numeric(x)) round(var(x, na.rm = TRUE), 6) else NA
})

sds <- sapply(covid_ned_selected, function(x) {
  if(is.numeric(x)) round(sd(x, na.rm = TRUE), 6) else NA
})

skewnesses <- sapply(covid_ned_selected, function(x) {
  if(is.numeric(x)) round(skewness(x, na.rm = TRUE), 6) else NA
})

kurtoses <- sapply(covid_ned_selected, function(x) {
  if(is.numeric(x)) round(kurtosis(x, na.rm = TRUE), 6) else NA
})

summary_table <- data.frame(
  Variable = names(modes),
  Mode = modes,
  Mean = means,
  Median = medians,
  Range = ranges,
  Variance = variances,
  Std_Dev = sds,
  Skewness = skewnesses,
  Kurtosis = kurtoses,
  Interpretation = explanations,
  stringsAsFactors = FALSE
)
```

```
kable(summary_table) %>%
  kable_styling(bootstrap_options=c("bordered", "responsive","striped"), full_width = FALSE)
```

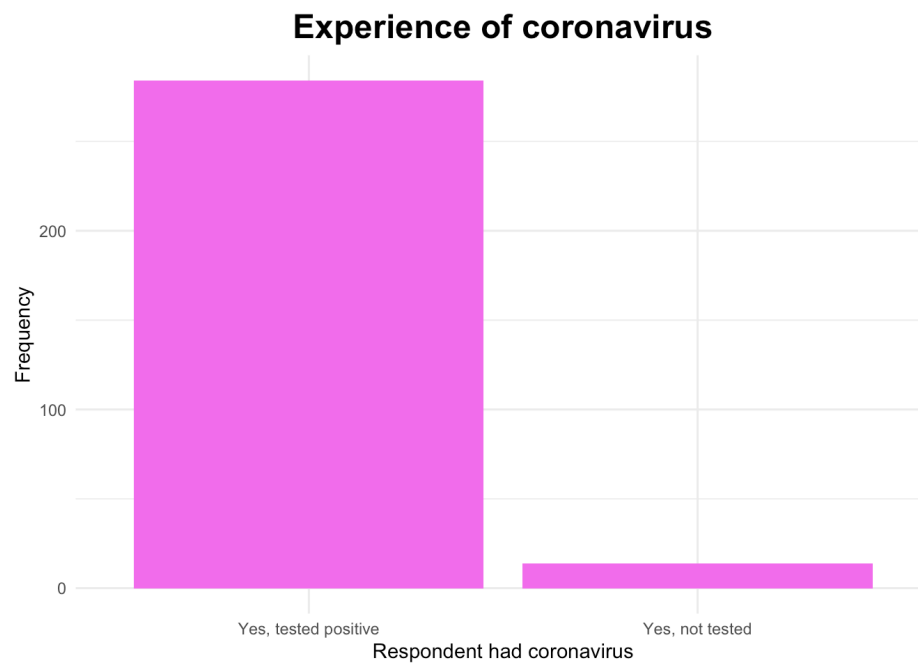
	Variable	Mode	Mean	Median	Range	Variance	Std_Dev	Skewness	Kurtosis	Interpretation
respc19a	respc19a	Yes, tested positive	NA	NA	NA	NA	NA	NA	NA	Mode indicates that most of respondents had coronavirus and tested positive.
symtc19	symtc19	Yes	NA	NA	NA	NA	NA	NA	NA	Mode "Yes" means that most of respondents had symptoms of coronavirus lasting 3 months or longer.
symtnc19	symtnc19	No	NA	NA	NA	NA	NA	NA	NA	Mode "No" means most of the respondents do not still have symptoms of coronavirus.
vacc19	vacc19	Yes	NA	NA	NA	NA	NA	NA	NA	Mode shows that most of the respondents received at least one dose of a vaccine.
psppsgva	psppsgva	Some	NA	NA	NA	NA	NA	NA	NA	The mode is 3, so most respondents chose the answer right in the middle - the political system in [country] allows people like them to have some say in what the government does.

	Variable	Mode	Mean	Median	Range	Variance	Std_Dev	Skewness	Kurtosis	Interpretation
stfhlth	stfhlth	8	6.308725	7	10	3.830294	1.957114	-0.925186	3.756218	The most frequent estimation of health services in country is 8 (10 is extremely good), but a significant amount of respondents estimate health services lower because the mean answer is 6 and median is 7.
stfgov	stfgov	5	4.577181	5	10	4.029377	2.007331	-0.413429	2.800726	Mode, mean and median are very close to each other, which means that data about satisfaction in government is quite normally distributed and most people are medium satisfied.
health	health	2	2.342282	2	4	0.596253	0.772174	0.294542	3.006398	Subjective health statistics are similar; health is good for most of the respondents.

Comment: We see that stfhlth and stfgov will be skewed to the left tail and health to the right. Kurtosis near 3 means that distribution of numeric variables is normal.

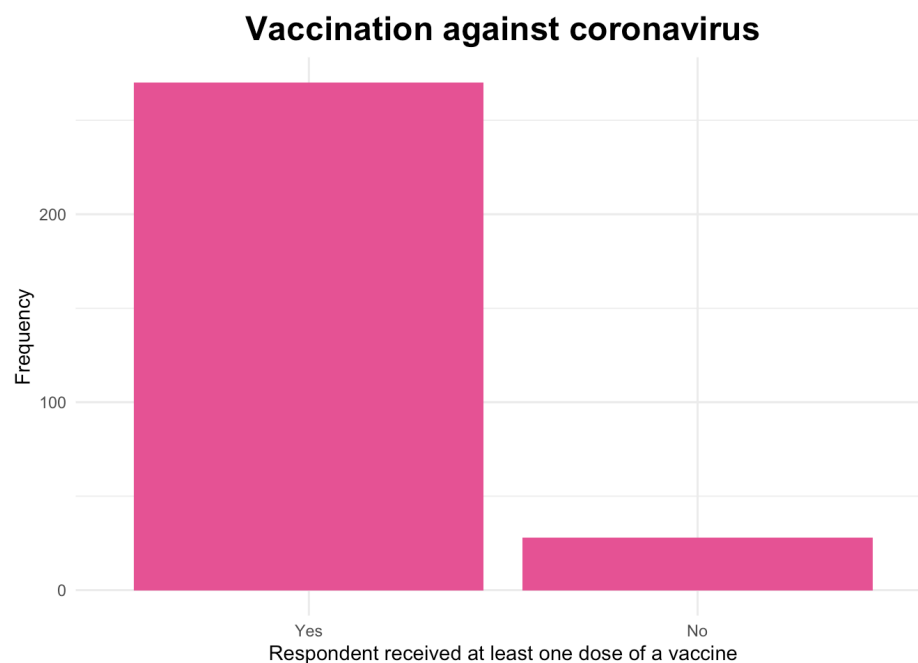
Bar plots

```
ggplot(covid_ned_selected, aes(x = respc19a)) +
  geom_bar(fill = "#f46fec") +
  theme_minimal() +
  labs(title = "Experience of coronavirus", x = "Respondent had coronavirus", y = "Frequency") +
  theme(plot.title = element_text(face = "bold", size = 18, hjust = 0.5))
```



Comment Bar plot shows that significantly more people had coronavirus and were tested positive for it than people who had not had coronavirus. The number of respondents who were not tested, but thought they had coronavirus is close to zero.

```
ggplot(covid_ned_selected, aes(x = vacc19)) +
  geom_bar(fill = "#e75396") +
  theme_minimal() +
  labs(title = "Vaccination against coronavirus", x = "Respondent received at least one dose of a vaccine", y = "Frequency") +
  theme(plot.title = element_text(face = "bold", size = 18, hjust = 0.5))
```

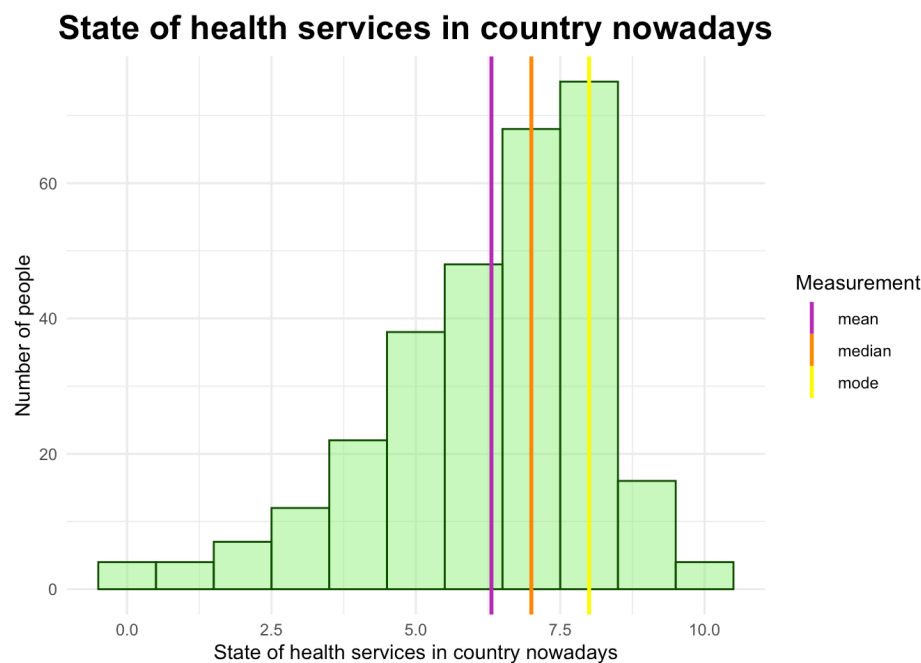


Comment This plot illustrates that most of the respondents had vaccinated against coronavirus.

Histograms

```
mode_val <- calculate_mode(covid_ned_selected$stfhlth)
mode_val <- as.numeric(as.character(mode_val))

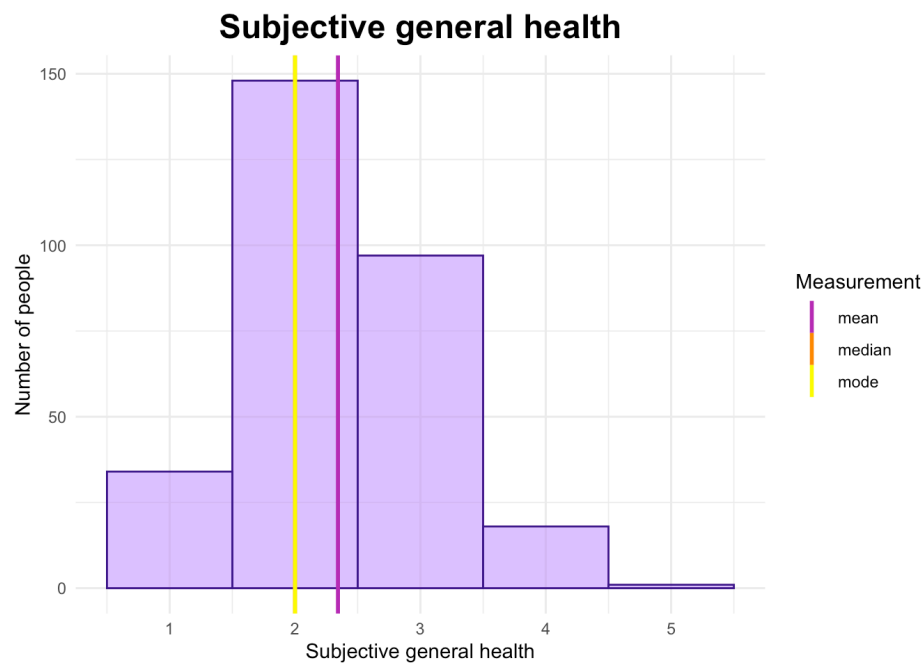
ggplot()+
  geom_histogram(data = covid_ned_selected, aes(x = stfhlth), binwidth = 1, fill="#8ef278", col="#0e5000", alpha = 0.5) +
  xlab("State of health services in country nowadays") +
  ylab("Number of people") +
  geom_vline(aes(xintercept = mean(covid_ned_selected$stfhlth), color = 'mean'), linetype="solid",
size=1) +
  geom_vline(aes(xintercept = median(covid_ned_selected$stfhlth), color = 'median'), linetype="solid",
size=1)+
  geom_vline(aes(xintercept = mode_val, color = 'mode'), linetype="solid",size=1) +
  scale_color_manual(name = "Measurement", values = c(median = "#ff8b00", mean = "#b716b4", mode =
"#fbff00"))+
  ggtitle("State of health services in country nowadays") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 18, hjust = 0.5))
```



Comment The level of perceived state of health services in Netherlands is skewed to the right. This means that while a majority perceive the health services positively, some individuals have significantly less favorable experiences, leading to the skewness. Most frequently given rating for the health services is 8 out of 10. 50% of the ratings are above 7 and 50% are below 7.

```
mode_val1 <- calculate_mode(covid_ned_selected$health)
mode_val1 <- as.numeric(as.character(mode_val1))

ggplot()+
  geom_histogram(data = covid_ned_selected, aes(x = health), binwidth = 1, fill="#bc83ff", col="#41008d", alpha = 0.5) +
  xlab("Subjective general health") +
  ylab("Number of people") +
  geom_vline(aes(xintercept = mean(covid_ned_selected$health), color = 'mean'), linetype="solid",
size=1) +
  geom_vline(aes(xintercept = median(covid_ned_selected$health), color = 'median'), linetype="solid",
size=1)+
  geom_vline(aes(xintercept = mode_val1, color = 'mode'), linetype="solid",size=1) +
  scale_color_manual(name = "Measurement", values = c(median = "#ff8b00", mean = "#b716b4", mode =
"#fbff00"))+
  ggtitle("Subjective general health") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 18, hjust = 0.5))
```



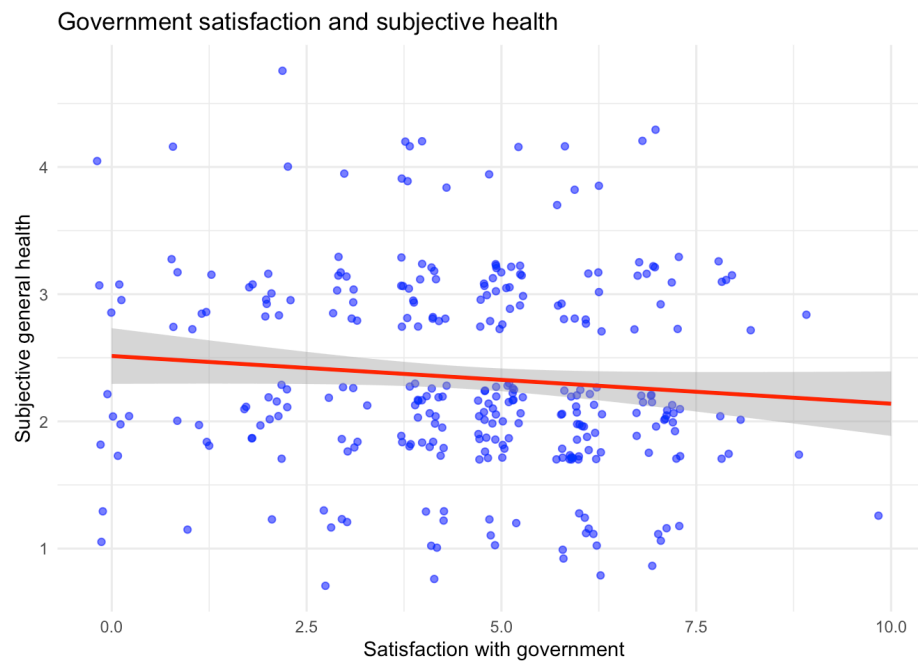
The most common rating given by individuals is 2, which suggests that many people perceive their health as good. The median also being 2 indicates that half of the respondents rated their health at or below this level. The histogram's left skewness also suggests that there are more ratings concentrated at the lower end of the scale, with fewer individuals providing higher ratings.

Scatterplot

Justification:

- The way governments handled COVID-19 has shaped long-term public trust in institutions, which may still influence how people evaluate their well-being and healthcare access.
- If people perceive these policies as beneficial, they may feel more secure and healthier, even if the direct effects of COVID-19 have faded.

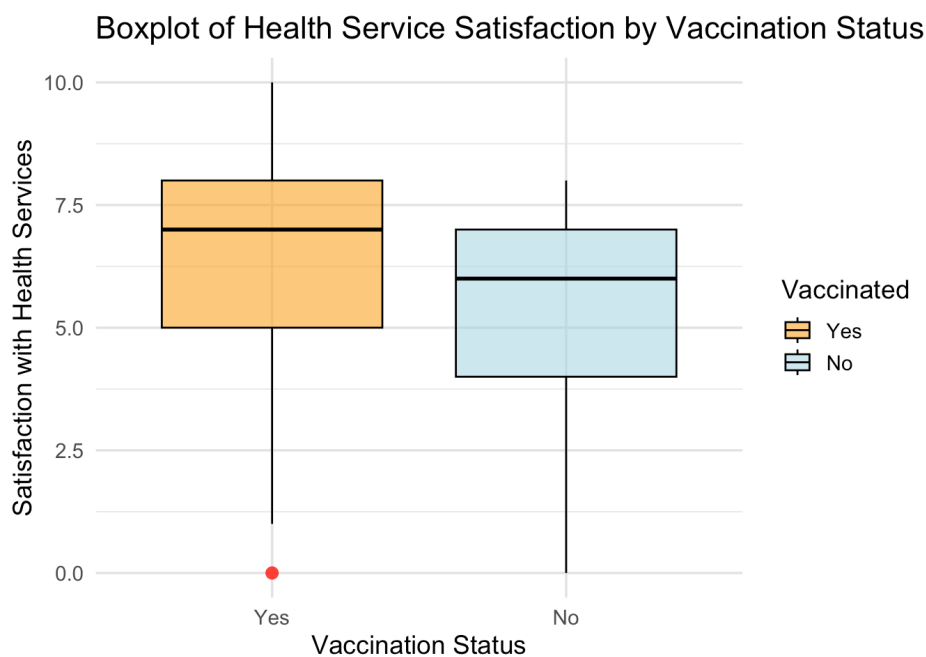
```
ggplot(covid_ned_selected, aes(x = stfgov, y = health)) +
  geom_jitter(width = 0.3, height = 0.3, alpha = 0.6, color = "blue") +
  geom_smooth(method = "lm", se = TRUE, color = "red") +
  labs(
    title = "Government satisfaction and subjective health",
    x = "Satisfaction with government",
    y = "Subjective general health"
  ) +
  theme_minimal()
```



Comment: There is a weak negative association between satisfaction with government and subjective health: people who are more satisfied with the government tend to report slightly better health, but the relationship is not strong and there is substantial variation in the data.

Boxplot

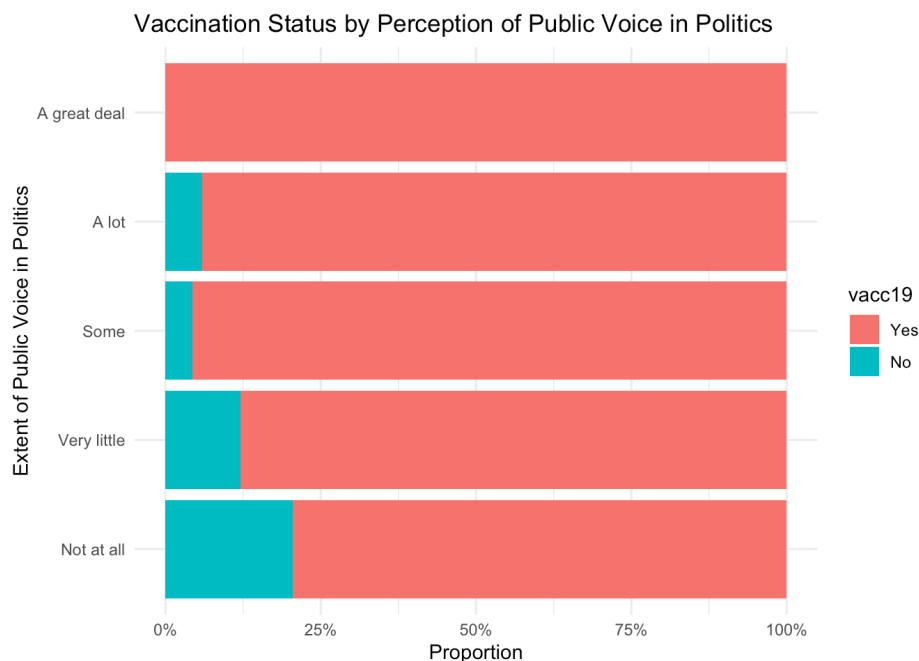
```
ggplot(covid_ned_selected, aes(x = vacc19, y = stfhlth, fill = vacc19)) +
  geom_boxplot(alpha = 0.6, color = "black", outlier.color = "red", outlier.shape = 16, outlier.size = 3) +
  scale_fill_manual(values = c("No" = "lightblue", "Yes" = "orange")) +
  labs(title = "Boxplot of Health Service Satisfaction by Vaccination Status",
       x = "Vaccination Status",
       y = "Satisfaction with Health Services",
       fill = "Vaccinated") +
  theme_minimal(base_size = 14) +
  theme(panel.grid.major = element_line(color = "gray90"))
```



Comment: This boxplot compares health service satisfaction by vaccination status, showing that the median satisfaction score is slightly higher for the unvaccinated group, though both groups have a similar interquartile range (IQR), indicating comparable variability. The vaccinated group exhibits a wider range of satisfaction scores, while the unvaccinated group has a more concentrated distribution with fewer extreme values. Additionally, both groups have outliers, but the unvaccinated group has more visible low-value outliers.

Stacked bar plot

```
ggplot(data = covid_ned_selected, aes(x = psppsgva, fill = vacc19)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  coord_flip() +
  xlab("Extent of Public Voice in Politics") +
  ylab("Proportion") +
  ggtitle("Vaccination Status by Perception of Public Voice in Politics") +
  theme_minimal()
```



Comment: Barplot shows that people who feel more involved and influential in politics tend to be vaccinated from coronavirus, while those who feel less influential are more likely to be unvaccinated. This suggests that high political engagement may be accompanied by greater trust in public institutions and official health measures, which is important to consider in context of relationship between trust in government and satisfaction with healthcare in the Netherlands. It is also important to note that the vaccination variable is highly unbalanced, which limits the variability of the data.

Conclusion

This research examined the extent to which trust in government institutions and satisfaction with the healthcare system are associated with the likelihood of COVID-19 vaccination in the Netherlands.

A major finding is that people who believe they have a bigger public voice in politics are more likely to get vaccinated, whereas those who believe they have little influence in political decision-making are less likely to be vaccinated. This suggests that political activity and trust of institutional authority may be associated with the tendency to receive a vaccine. Additionally, healthcare satisfaction does not strongly differentiate between vaccinated and unvaccinated individuals, though the unvaccinated group appears to have a slightly higher median satisfaction score and a more concentrated distribution of responses.

Overall, our findings highlight the complex connection between institutional trust, political perceptions and healthcare satisfaction in shaping public health behaviors. While healthcare satisfaction alone does not appear to be a decisive factor in vaccination decisions, attitudes toward government and political influence emerge as key drivers of vaccine hesitancy.

References

van Dullemen, R., & de Bruijn, H. (2021). The Dutch approach to COVID-19: How is it distinctive? NDU Press. Retrieved from <https://ndupress.ndu.edu/Media/News/News-Article-View/Article/2920148/the-dutch-approach-to-covid-19-how-is-it-distinctive/> (<https://ndupress.ndu.edu/Media/News/News-Article-View/Article/2920148/the-dutch-approach-to-covid-19-how-is-it-distinctive/>)

Yousuf H, van der Linden S, van Essen T, et al. Dutch Perspectives Toward Governmental Trust, Vaccination, Myths, and Knowledge About Vaccines and COVID-19. JAMA Netw Open. 2021;4(12):e2140529. doi:10.1001/jamanetworkopen.2021.40529 (doi:10.1001/jamanetworkopen.2021.40529)

PROJECT 2

Topic

Public Perceptions and Responses to COVID-19 in Netherlands

Research Question

To what extent do COVID-19 experiences and vaccination status influence political trust and civic engagement in the Netherlands?

Relevance and justification

The COVID-19 epidemic has had a major effect on public opinions of political institutions and civic practices around the world, including the Netherlands. The method to handling the epidemic, which began with an “intelligent lockdown,” evolved over time, balancing public health precautions with economic concerns. This method had an impact on public trust levels, as the population’s emotional response remained overwhelmingly positive, expressing trust in government actions during the early stages of the pandemic. However, the epidemic exacerbated pre-existing social conflicts. In recent years, the Netherlands has seen numerous demonstrations, demonstrating both active civic engagement and underlying public dissatisfaction. The government recognizes these feelings and is working to restore faith in public institutions, acknowledging that the pandemic has contributed to existing issues in education, healthcare, and employment (U.N. Human Rights Council, 2022).

Data loading and selection

```
covid_ned_selected <- dplyr::select(covid_ned, respc19a, trstplt, vacc19, sgnptit)

covid_ned_selected <- covid_ned_selected %>%
  mutate(respc19a = ifelse(respc19a %in% c(7, 8, 9), NA, respc19a)) %>%
  drop_na(respc19a)

covid_ned_selected <- covid_ned_selected %>%
  mutate(
    vacc19 = ifelse(vacc19 %in% c(7, 8), NA, vacc19),
    trstplt = ifelse(trstplt %in% c(88), NA, trstplt),
    respc19a = ifelse(respc19a %in% c(8), NA, respc19a),
    sgnptit = ifelse(sgnptit %in% c(8), NA, sgnptit)
  ) %>%
  drop_na()

covid_ned_selected <- covid_ned_selected %>%
  mutate(vacc19 = ifelse(vacc19 %in% c(7, 8, 9), NA, vacc19),
    respc19a = ifelse(respc19a %in% c(7, 8, 9), NA, respc19a),
    trstplt = ifelse(trstplt %in% c(77, 88, 99), NA, trstplt),
    sgnptit = ifelse(sgnptit %in% c(7, 8, 9), NA, sgnptit)) %>%
  drop_na()
```

Chi-squared test

```
Variable_names <- c("vacc19", "sgnptit")
Meanings <- c("Respondent received at least one dose of a vaccine against coronavirus approved by the national regulatory authority in [country]?", "Signed petition last 12 months")
Types <- c("Nominal", "Nominal")

df <- data.frame(Variable_names, Meanings, Types, stringsAsFactors = FALSE)
kable(df) %>%
  kable_styling(bootstrap_options=c("bordered", "responsive", "striped"), full_width = FALSE)
```

Variable_names	Meanings	Types
vacc19	Respondent received at least one dose of a vaccine against coronavirus approved by the national regulatory authority in [country]?	Nominal

Variable_names	Meanings	Types
sgnptit	Signed petition last 12 months	Nominal

Hypotheses:

H_0 : Vaccination status does not differ by petition response.

H_1 : Vaccination status differs by petition response.

Justification:

Trust in government and political engagement are important factors influencing public health behaviors, including vaccination. Individuals who participate in civic actions such as signing petitions may be more politically aware and trusting of institutional authority. Research has shown that higher trust in government is associated with greater willingness to receive vaccines (Gollust et al., 2020). Moreover, political participation has been linked to compliance with public health measures (Jennings et al., 2021). Therefore, it is plausible that individuals who signed a petition in the past year may be more likely to have received a COVID-19 vaccine, reflecting a broader orientation toward civic responsibility and trust in public institutions.

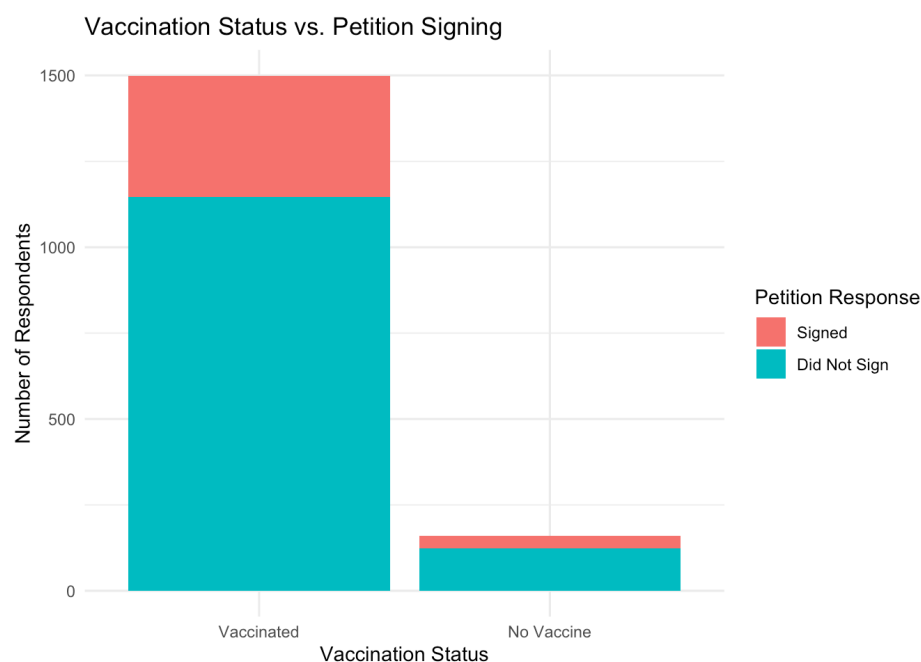
Converting variables

```
data_clean <- covid_ned_selected %>%
  mutate(vacc19 = as.factor(vacc19),
         sgnptit = as.factor(sgnptit))
```

Stacked bar plot

```
data_clean$vacc19_label <- factor(data_clean$vacc19, levels = c(1, 2), labels = c("Vaccinated", "No Vaccine"))
data_clean$sgnptit_label <- factor(data_clean$sgnptit, levels = c(1, 2), labels = c("Signed", "Did Not Sign"))

ggplot(data_clean, aes(x = vacc19_label, fill = sgnptit_label)) +
  geom_bar(position = "stack") +
  labs(
    title = "Vaccination Status vs. Petition Signing",
    x = "Vaccination Status",
    y = "Number of Respondents",
    fill = "Petition Response"
  ) +
  theme_minimal()
```



Creating contingency table

```
table_data <- table(data_clean$ vacc19, data_clean$sgnptit)
print(table_data)
```

```
##
##      1      2
## 1  353 1146
## 2   38  123
```

Values are whole numbers, these are counts -> assumption met.

```
str(data_clean$ vacc19)
```

```
## Factor w/ 2 levels "1","2": 1 1 1 1 1 1 1 1 1 1 ...
```

```
str(data_clean$sgnptit)
```

```
## Factor w/ 2 levels "1","2": 2 2 2 2 2 1 1 2 1 2 ...
```

Type is "factor" -> assumption is met.

```
chi_test1 <- chisq.test(table_data)
print(chi_test1)
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  table_data
## X-squared = 2.8197e-30, df = 1, p-value = 1
```

There is no significant association between vaccination status and petition signing ($p = 1$). The observed frequencies are almost identical to the expected frequencies, meaning the variables appear to be completely independent.

```
print(chi_test1$expected)
```

```
##
##      1      2
## 1 353.07771 1145.9223
## 2  37.92229  123.0777
```

All expected cell counts are ≥ 5 -> assumption is met.

```
std_resid <- chi_test1$stdres
print(round(std_resid, 2))
```

```
##
##      1      2
## 1 -0.02  0.02
## 2  0.02 -0.02
```

None of the residuals exceeded ± 2 , indicating no meaningful deviation between observed and expected frequencies in any cell. This supports the Chi-square result, which was not statistically significant, suggesting no evidence of association between receiving a COVID-19 vaccine and signing a petition in the past 12 months.

T-test

```
Variable_names <- c("vacc19", "trstplt")
Meanings <- c("Respondent received at least one dose of a vaccine against coronavirus approved by the national regulatory authority in [country]?", "Trust in politicians")
Types <- c("Nominal", "Quasi-interval")

df <- data.frame(Variable_names, Meanings, Types, stringsAsFactors = FALSE)
kable(df) %>%
  kable_styling(bootstrap_options=c("bordered", "responsive","striped"), full_width = FALSE)
```

Variable_names	Meanings	Types
vacc19	Respondent received at least one dose of a vaccine against coronavirus approved by the national regulatory authority in [country]?	Nominal
trstplt	Trust in politicians	Quasi-interval

Check Data Types for Selected Variables

```
class(covid_ned_selected$vacc19)
```

```
## [1] "integer"
```

```
class(covid_ned_selected$trstplt)
```

```
## [1] "integer"
```

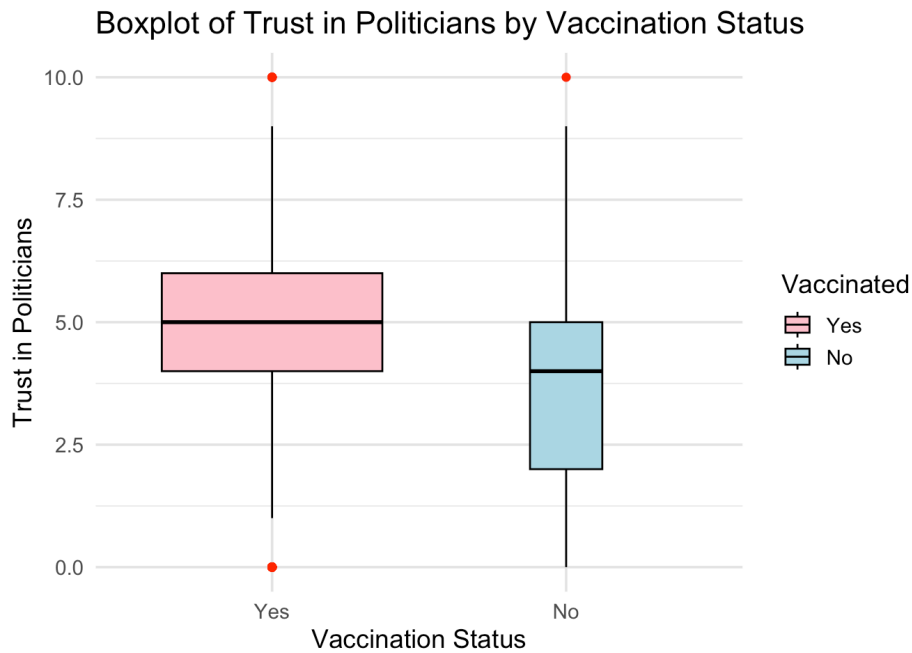
Conversion of Variable's Type

A t-test is used to compare the means of a numeric variable between two groups that are defined by a categorical variable. In this case, we need to treat vacc19 as a categorical variable.

```
covid_ned_selected$vacc19 <- factor(covid_ned_selected$vacc19,
                                   levels = c(1, 2),
                                   labels = c("Yes", "No"))
```

Boxplot

```
ggplot(covid_ned_selected, aes(x = factor(vacc19), y = as.numeric(trstplt), fill = factor(vacc19))) +
  geom_boxplot(varwidth = TRUE, alpha = 1, color = "black", outlier.color = "red", outlier.shape = 16, outlier.size = 2) +
  scale_fill_manual(values = c("No" = "lightblue", "Yes" = "pink")) +
  labs(title = "Boxplot of Trust in Politicians by Vaccination Status",
       x = "Vaccination Status",
       y = "Trust in Politicians",
       fill = "Vaccinated") +
  theme_minimal(base_size = 14) +
  theme(panel.grid.major = element_line(color = "gray90"))
```



Comment:

- The median trust in politicians is higher among the vaccinated group than among the unvaccinated group.
- The vaccinated group has a wider IQR, indicating greater variability in trust levels within the middle 50% of responses.
- The IQR for the vaccinated group is narrower, indicating that their trust in politicians is more tightly clustered around the median. In contrast, the unvaccinated group has a wider IQR, showing greater variability in trust levels.
- Whiskers extend from approximately 0 to 10 in both groups, showing a similar overall range.
- Both groups have outliers at both high and low trust levels, indicating some extreme responses in both categories.

Normality Check

1) Descriptive Statistics

```
describeBy(covid_ned_selected$trstplt, covid_ned_selected$vacc19)
```

```
##
## Descriptive statistics by group
## group: Yes
##   vars   n mean   sd median trimmed  mad min max range skew kurtosis   se
## X1     1 1499 5.06 1.98      5    5.24 1.48   0  10   10 -0.7    0.13 0.05
## -----
## group: No
##   vars   n mean   sd median trimmed  mad min max range skew kurtosis   se
## X1     1  161  3.8 2.34      4    3.82 2.97   0  10   10 -0.19  -0.85 0.18
```

```
table(covid_ned_selected$vacc19)
```

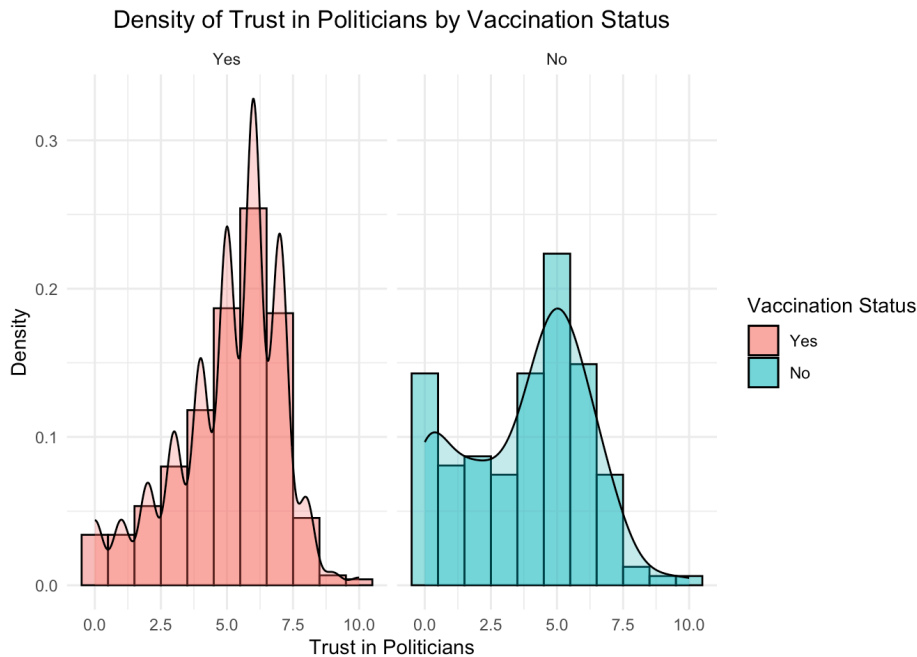
```
##
##   Yes   No
## 1499  161
```

Comment:

- Mean and median in both cases are close to each other.
- The first category has skew = -0.7 representing a moderate left skew. Second one is close to 0.
- The first category has kurtosis close to normal (0), while the second one has kurtosis = -0.85, meaning a flatter distribution.
- The second category has more variability in trust in politicians (SD = 2.34).
- **1st category:** The left skew suggests that the distribution is not perfectly normal, despite the mean and median being close and the kurtosis being close to 0.
- **2nd category:** The skewness close to 0 and corresponds to normality, but the kurtosis of -0.85 suggests a flatter distribution, which deviates from a normal distribution.

2) Histogram

```
ggplot(covid_ned_selected, aes(x = trstplt, fill = factor(vacc19))) +
  geom_histogram(aes(y = after_stat(density)), binwidth = 1, color = "black", alpha = 0.5) +
  geom_density(alpha = 0.3) +
  facet_wrap(~ vacc19) +
  labs(title = "Density of Trust in Politicians by Vaccination Status",
       x = "Trust in Politicians",
       y = "Density",
       fill = "Vaccination Status") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))
```



Comment:

For vaccinated group:

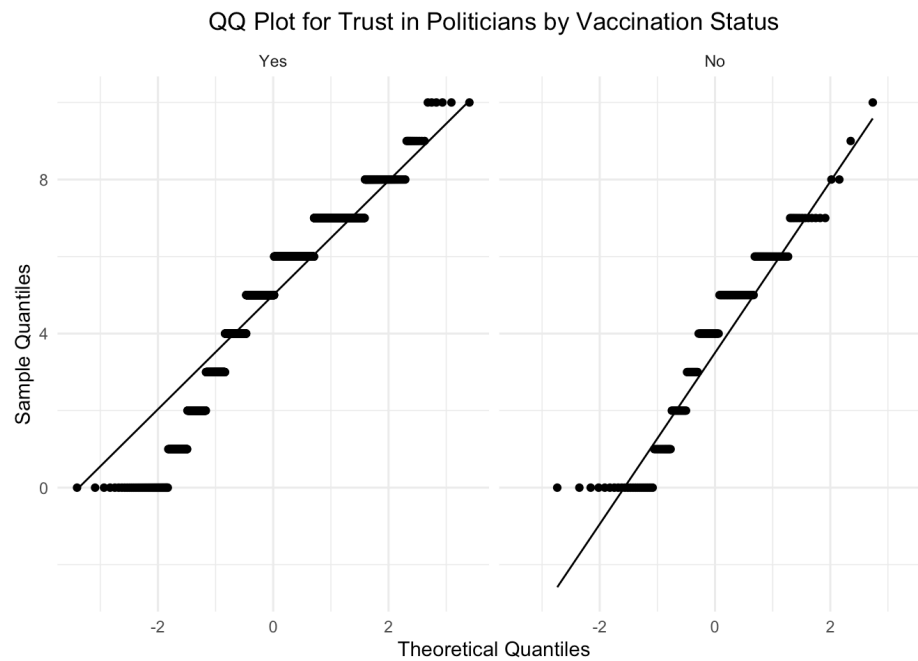
- Vaccinated individuals tend to have a higher and more concentrated level of trust in politicians.
- The distribution appears roughly bell-shaped, with a peak around 6 and a gradual narrowing towards both ends.
- However, it shows a slight right skew, meaning there are more values concentrated on the lower end, but some higher trust scores still exist.
- The shape looks like a normal distribution but may not be perfectly normal.

For unvaccinated group:

- Unvaccinated individuals show a much lower frequency overall, suggesting fewer respondents in this category.
- The distribution is much flatter and more dispersed, lacking a clear central peak.
- The spread is more uniform, and there is no strong indication of a normal shape, suggesting it is a non-normal distribution.

3) Q-Q Plot

```
ggplot(covid_ned_selected, aes(sample = trstplt)) +
  stat_qq() +
  stat_qq_line() +
  labs(title = "QQ Plot for Trust in Politicians by Vaccination Status",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles") +
  theme_minimal() +
  facet_wrap(~vacc19) +
  theme(plot.title = element_text(hjust = 0.5))
```



Comment:

- Both vaccinated and unvaccinated groups show similar deviations from normality.
- In both groups, we see a pattern where data points at the lower and upper ends deviate from the line.
- Both groups exhibit right-skewed distributions.
- However, the vaccinated group appears to have stronger skewness.
- As our variable is a scale from 0 to 10 (ordinal) it is not continuous so there are clear steps between data points.

4) Shapiro-Wilk Test

H_0 : The data follows a normal distribution.

H_1 : The data does not follow a normal distribution.

```
shapiro.test(covid_ned_selected$trstplt[covid_ned_selected$vacc19 == "No"])
```

```
##
##  Shapiro-Wilk normality test
##
## data:  covid_ned_selected$trstplt[covid_ned_selected$vacc19 == "No"]
## W = 0.93295, p-value = 7.439e-07
```

```
shapiro.test(covid_ned_selected$trstplt[covid_ned_selected$vacc19 == "Yes"])
```

```
##
##  Shapiro-Wilk normality test
##
## data:  covid_ned_selected$trstplt[covid_ned_selected$vacc19 == "Yes"]
## W = 0.93173, p-value < 2.2e-16
```

Comment: In both cases p-value < 0.05, so we reject the null hypothesis and derive the conclusion that data not normally distributed in two categories.

Equality of Variances

Formal test for homogeneity of variances.

H_0 : The variances are equal between the two categories.

H_1 : The variances are not equal between the two categories.

```
leveneTest(covid_ned_selected$trstplt ~ covid_ned_selected$vacc19)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value    Pr(>F)
## group  1 13.357 0.0002655 ***
##      1658
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Comment: p-value < 0.05, so we reject the null hypothesis. Two categories have significantly different variances.

Hypotheses:

H_0 : There is no difference in the mean trust in politicians between vaccinated and unvaccinated groups.

H_1 : Vaccinated individuals have higher trust in politicians than unvaccinated individuals.

Justification:

Research shows that trust in politicians is a key determinant of people's health behaviors, including vaccination. Vaccinated people are more likely to trust politicians as they are more likely to follow government health recommendations and view political leaders as trustworthy sources of information. Trust in government has been found to be positively correlated with the number of vaccinations (Gollust et al., 2020), as people who perceive government as transparent and responsible are more willing to comply with health policies.

One-tailed Student's T-test

We conduct it just for the task as normality and homoscedasticity are violated.

```
t.test(covid_ned_selected$trstplt[covid_ned_selected$vacc19 == "Yes"],
       covid_ned_selected$trstplt[covid_ned_selected$vacc19 == "No"],
       alternative = "greater")
```

```
##
## Welch Two Sample t-test
##
## data: covid_ned_selected$trstplt[covid_ned_selected$vacc19 == "Yes"] and covid_ned_selected$trstplt[covid_ned_selected$vacc19 == "No"]
## t = 6.5621, df = 185.3, p-value = 2.58e-10
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  0.9411812      Inf
## sample estimates:
## mean of x mean of y
##  5.059373  3.801242
```

```
# to calculate effect size of difference
cohen.d(covid_ned_selected$trstplt ~ covid_ned_selected$vacc19)
```

```
##
## Cohen's d
##
## d estimate: 0.6237572 (medium)
## 95 percent confidence interval:
##   lower   upper
## 0.4597080 0.7878064
```

Comment: p-value < 0.05, so we reject the null hypothesis and conclude that vaccinated people have significantly higher trust in politicians than unvaccinated ones. There is a moderate effect size.

Welch's T-test (Parametric)

This one we conduct as our variances are not equal, but due to non-normal distribution we should conduct non-parametric test.

```
covid_ned_selected$trstplt <- as.numeric(covid_ned_selected$trstplt)
t.test(covid_ned_selected$trstplt ~ covid_ned_selected$vacc19, data = covid_ned_selected, var.equal = FALSE, alternative = "greater")
```

```
##
## Welch Two Sample t-test
##
## data: covid_ned_selected$trstplt by covid_ned_selected$vacc19
## t = 6.5621, df = 185.3, p-value = 2.58e-10
## alternative hypothesis: true difference in means between group Yes and group No is greater than 0
## 95 percent confidence interval:
## 0.9411812      Inf
## sample estimates:
## mean in group Yes mean in group No
##      5.059373      3.801242
```

```
cohens_d(covid_ned_selected, trstplt ~ vacc19, var.equal = FALSE)
```

```
## # A tibble: 1 × 7
##   .y.      group1 group2 effsize    n1    n2 magnitude
## * <chr>   <chr>   <chr>   <dbl> <int> <int> <ord>
## 1 trstplt Yes     No      0.580  1499   161 moderate
```

Comment: p-value < 0.05, so we reject the null hypothesis and conclude that vaccinated people have significantly higher trust in politicians than unvaccinated ones. There is a moderate effect size.

Wilcoxon Test (Non-parametric)

Finally, here is the test that will be relevant in our case.

```
wilcox.test(covid_ned_selected$trstplt ~ covid_ned_selected$vacc19, data = covid_ned_selected, alternative = "greater")
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: covid_ned_selected$trstplt by covid_ned_selected$vacc19
## W = 159600, p-value = 4.061e-12
## alternative hypothesis: true location shift is greater than 0
```

Comment: p-value < 0.05, which means that there is a highly significant difference between the two groups in terms of trust in politicians, with the vaccinated group showing higher trust in politicians.

Effect Size

```
rstatix::wilcox_effsize(trstplt ~ vacc19, data = covid_ned_selected)
```

```
## # A tibble: 1 × 7
##   .y.      group1 group2 effsize    n1    n2 magnitude
## * <chr>   <chr>   <chr>   <dbl> <int> <int> <ord>
## 1 trstplt Yes     No      0.168  1499   161 small
```

Comment: The calculated effect size for the difference in trust in politicians between vaccinated and unvaccinated groups is small according to Cohen's d criteria.

Hedges's g

When comparing groups with unequal sample sizes as in our case with 161 and 1499 observations, the traditional Cohen's d can be biased. Thus, Hedges' g correction is applied to adjust for the bias introduced by unequal sample sizes.

```
covid_ned_selected$vacc19 <- as.factor(covid_ned_selected$vacc19)
cohen.d(covid_ned_selected$trstplt ~ covid_ned_selected$vacc19, hedges.correction = TRUE)
```

```
##
## Hedges's g
##
## g estimate: 0.623475 (medium)
## 95 percent confidence interval:
##      lower      upper
## 0.4595013 0.7874488
```

Comment: The effect size of 0.6 means that vaccinated individuals tend to have a moderately higher level of trust in politicians compared to unvaccinated individuals. Also the entire confidence interval is above 0.2 confirms that there is a moderate to large effect, suggesting that vaccination status has a meaningful impact on trust in politicians.

ANOVA

Filtrating the data

```
covid_ned_selected <- covid_ned_selected %>%
  mutate(respc19a =
    case_when(
      respc19a == 1 ~ "Yes, I tested positive for coronavirus",
      respc19a == 2 ~ "Yes, I think I had coronavirus but was not tested/did not test positive",
      respc19a == 3 ~ "No, I have not had coronavirus",
      TRUE ~ as.character(NA) # Handle other cases (if any) as NA
    ) %>%
    drop_na(respc19a)

covid_ned_selected$trstplt <- as.numeric(as.character(covid_ned_selected$trstplt))
covid_ned_selected$respc19a <- as.factor(covid_ned_selected$respc19a)
```

Describing variables

```
variable_names <- c("respc19a", "trstplt")
meanings <- c("Respondent had coronavirus", "Trust in politicians")
measurements <- c("Yes and tested positive - Yes but not tested/did not test positive - No", "1 to 10 scale")
types <- c("Nominal", "Interval")

table <- data.frame(
  Variable = variable_names,
  Meaning = meanings,
  Type = types,
  Measurement = measurements,
  stringsAsFactors = FALSE
)

kable(table) %>%
  kable_styling(bootstrap_options=c("bordered", "stripped"), full_width = FALSE)
```

Variable	Meaning	Type	Measurement
respc19a	Respondent had coronavirus	Nominal	Yes and tested positive - Yes but not tested/did not test positive - No
trstplt	Trust in politicians	Interval	1 to 10 scale

We chose respc19a as independent categorical variable and trstplt as dependent continuous variable.

Descriptives across the groups

```
describeBy(covid_ned_selected$trstplt, covid_ned_selected$respc19a, mat = TRUE) %>%
  dplyr::rename(
    respc19a = group1,
    N = n,
    Mean = mean,
    SD = sd,
    Median = median,
    Min = min,
    Max = max,
    Skew = skew,
    Kurtosis = kurtosis,
    st.error = se
  ) %>%
  dplyr::select(respc19a, N, Mean, SD, Median, Min, Max, Skew, Kurtosis, st.error) %>%
  kable(
    align = c("l", rep("r", 9)),
    digits = 2,
    row.names = FALSE,
    caption = "The level of trust in politicians"
  ) %>%
  kable_styling(bootstrap_options = c("bordered", "striped"), full_width = FALSE)
```

The level of trust in politicians

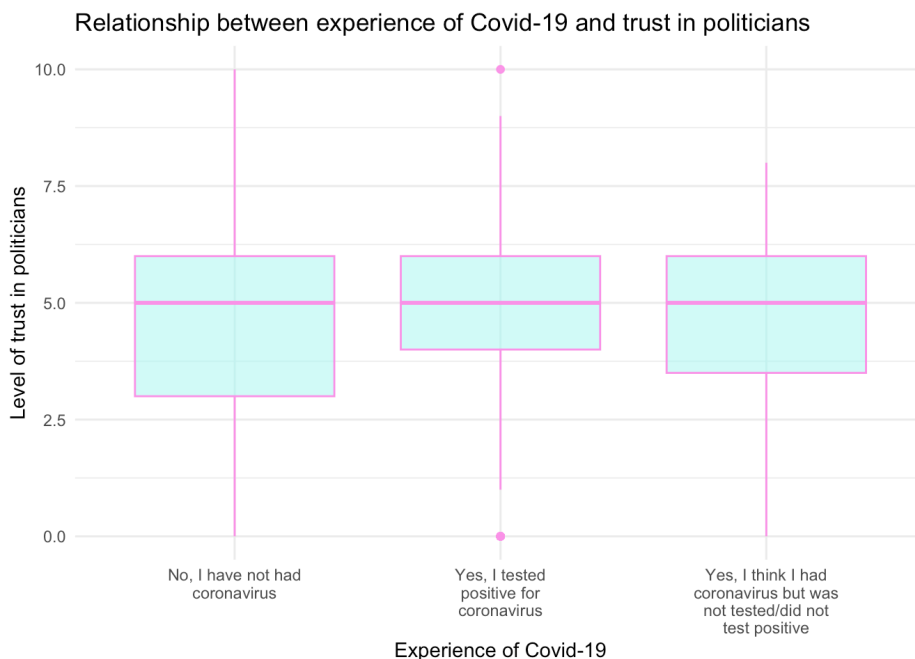
respc19a	N	Mean	SD	Median	Min	Max	Skew	Kurtosis	st.error
No, I have not had coronavirus	425	4.70	2.19	5	0	10	-0.43	-0.35	0.11
Yes, I tested positive for coronavirus	1172	5.04	1.98	5	0	10	-0.75	0.17	0.06
Yes, I think I had coronavirus but was not tested/did not test positive	63	4.56	2.27	5	0	8	-0.79	-0.62	0.29

- **Sample Size (N):** The sample sizes differ considerably across the groups. The “Yes, I tested positive for coronavirus” group is the largest (N = 1175), followed by the “No, I have not had coronavirus” group (N = 425), and the “Yes, I think I had coronavirus but was not tested/did not test positive” group is the smallest (N = 63). The results of one-way ANOVA can be biased because of this disbalance across groups.
- **Mean:** The mean trust level is slightly higher for those who tested positive for coronavirus (5.04) compared to those who didn’t (4.70) or think they had it but weren’t tested (4.56). However, the differences seem small.
- **Median:** The median trust level is 5 for all three groups, suggesting that the “typical” respondent in each group has a similar level of trust.
- **Standard Deviation (SD):** The standard deviations are relatively similar across the groups (ranging from 1.98 to 2.27), indicating comparable variability within each group.
- **Skewness:** All groups show negative skewness. This means the distribution of trust scores leans slightly towards the higher end of the scale (more people with scores above the mean than below).
- **Kurtosis:** The kurtosis values vary. The “tested positive” group has a kurtosis of 0.16, indicating a slightly peaked distribution compared to a normal distribution. The other two groups have negative kurtosis values, suggesting flatter distributions than a normal distribution.

Boxplot

```
ggplot(covid_ned_selected, aes(x = respc19a, y = trstplt)) +
  geom_boxplot(fill="#a2faf6", col="#fc92e9", alpha = 0.5) +
  ylim(c(0,10)) +
  xlab("Experience of Covid-19") +
  ylab("Level of trust in politicians") +
  ggtitle("Relationship between experience of Covid-19 and trust in politicians") +
  theme(
    plot.title = element_text(hjust = 0.5),

    plot.margin = unit(c(1, 1, 1, 1), "cm"),
    axis.text.x = element_text(lineheight = 0.8),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank()
  ) +
  theme_minimal() +
  scale_x_discrete(labels = function(x) str_wrap(x, width = 20))
```



Interpretation: Based on the boxplot, the median trust levels are the same across all three groups, the means are close to center, the distribution of the data is also similar, and there are only two outliers in the "Yes, I tested positive for coronavirus" group. Therefore, Boxplot suggests that the experience of COVID-19 appears to have little effect on the level of trust in politicians.

Checking the homogeneity of variances

H0: Variances across groups are equal

```
leveneTest(trstplt ~ respc19a, data = covid_ned_selected)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group  2  3.3419 0.03561 *
##      1657
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation: The variances are unequal across the three groups defined by COVID-19 experience. This suggests that the spread of trust in politicians varies between groups, and the assumption of equal variances is violated. It means that one-way anova results may be unreliable so we will double-check it with non-parametric test.

Conducting One-way ANOVA

Hypotheses:

H_0 : The mean levels of trust in politicians across the between the groups defined by COVID-19 experience are equal. H_1 : The mean levels of trust in politicians differ between at least two groups defined by COVID-19 experience.

Justification:

COVID-19 has been a profound social and political event, and individuals' experiences with the virus (e.g., being infected, knowing someone who was infected or died, or having no direct experience) may influence their attitudes toward political authorities. Research suggests that personal experiences with public health crises can shape political trust, especially when individuals evaluate the government's crisis response (Devine et al., 2020). For example, those who perceive the government's pandemic response as inadequate may show lower trust, while those with positive or less direct experiences may maintain or even increase their trust in politicians.

```
oneway.test(trstplt ~ respc19a, data = covid_ned_selected, var.equal = T)
```

```
##  
## One-way analysis of means  
##  
## data: trstplt and respc19a  
## F = 5.4297, num df = 2, denom df = 1657, p-value = 0.004463
```

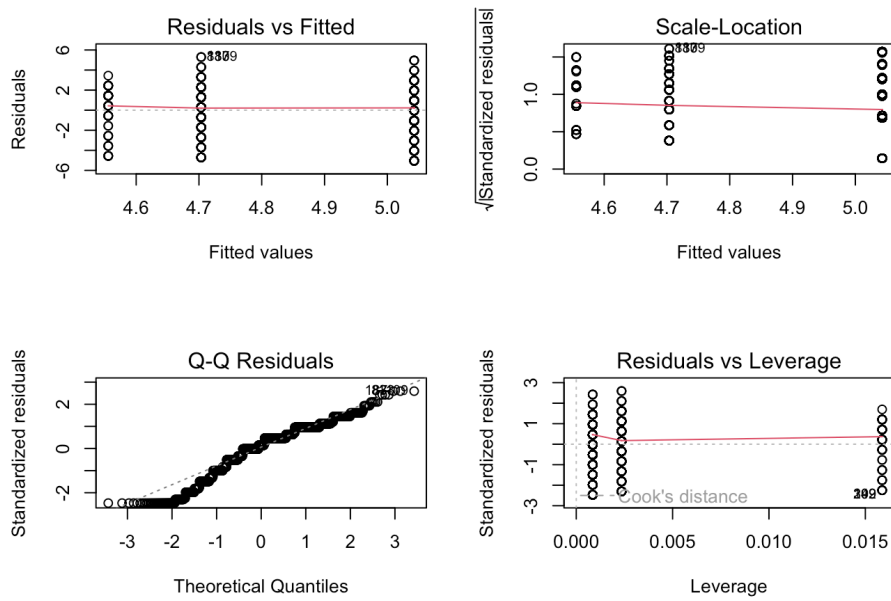
```
aov <- aov(trstplt ~ respc19a, data = covid_ned_selected)  
summary(aov)
```

```
##           Df Sum Sq Mean Sq F value    Pr(>F)  
## respc19a     2     45   22.709     5.43 0.00446 **  
## Residuals 1657   6930    4.182  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation: P-value < 0.05 and F is 5.43 which is quite big so H_0 can be rejected. There is significant difference in trust in politicians between the groups defined by COVID-19 experience. At least one group has level of trust in politicians which differs from other groups.

Checking for normality of residuals

```
layout(matrix(1:4, 2, 2))  
plot(aov)
```



Interpretation: The redline is quite straight on each graph but Q-Q plot shows some deviations from diagonal and several outliers so we can conclude that the distribution of residuals is not normal but still we will double-check the results with non-parametric test.

Post-Hoc test

```
TukeyHSD(aov)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = trstplt ~ respc19a, data = covid_ned_selected)
##
## $respc19a
##
diff
## Yes, I tested positive for coronavirus-No, I have not had coronavirus
0.3391327
## Yes, I think I had coronavirus but was not tested/did not test positive-No, I have not had coron
avirus -0.1479739
## Yes, I think I had coronavirus but was not tested/did not test positive-Yes, I tested positive f
or coronavirus -0.4871066
##
lwr
## Yes, I tested positive for coronavirus-No, I have not had coronavirus
0.06749192
## Yes, I think I had coronavirus but was not tested/did not test positive-No, I have not had coron
avirus -0.79563219
## Yes, I think I had coronavirus but was not tested/did not test positive-Yes, I tested positive f
or coronavirus -1.10754713
##
upr
## Yes, I tested positive for coronavirus-No, I have not had coronavirus
0.6107735
## Yes, I think I had coronavirus but was not tested/did not test positive-No, I have not had coron
avirus 0.4996845
## Yes, I think I had coronavirus but was not tested/did not test positive-Yes, I tested positive f
or coronavirus 0.1333340
##
p adj
## Yes, I tested positive for coronavirus-No, I have not had coronavirus
0.0096595
## Yes, I think I had coronavirus but was not tested/did not test positive-No, I have not had coron
avirus 0.8536279
## Yes, I think I had coronavirus but was not tested/did not test positive-Yes, I tested positive f
or coronavirus 0.1564339
```

Interpretation: Tukey shows that one pair has statistically significant difference 0.3391327 - Yes, I tested positive for coronavirus and No, I have not had coronavirus.

Effect size

```
eta_squared <- aov %>%
  eta_squared()
print(eta_squared)
```

```
## respc19a
## 0.006510972
```

Interpretation: Effect size is very small which means that Covid-19 experience explains very small part of trust in politicians variation.

Kruskal-Wallis test

H0: The mean levels of trust in politicians across the between the groups defined by COVID-19 experience are equal.

```
kruskal.test(trstplt ~ respc19a, data = covid_ned_selected)
```

```
##
## Kruskal-Wallis rank sum test
##
## data: trstplt by respc19a
## Kruskal-Wallis chi-squared = 9.6109, df = 2, p-value = 0.008185
```

Interpretation: The Kruskal-Wallis test confirms the results of one-way ANOVA. The p-value < 0.05 and a chi-squared is 9.6109 indicating a significant difference between the observed frequencies and the expected frequencies under a null hypothesis.

Dunn's test

```
dunn_test(trstplt ~ respc19a, data = covid_ned_selected)
```

```
## # A tibble: 3 × 9
##   .y.      group1      group2    n1    n2 statistic      p  p.adj p.adj.signif
## * <chr>    <chr>      <chr> <int> <int>    <dbl>    <dbl> <dbl> <chr>
## 1 trstplt No, I have n... Yes, ...   425   1172    2.93    0.00341 0.0102 *
## 2 trstplt No, I have n... Yes, ...   425    63   -0.0667 0.947    0.947 ns
## 3 trstplt Yes, I teste... Yes, ...  1172    63   -1.35    0.176    0.353 ns
```

Interpretation: Dunn test also confirms previous results. Only Yes, I tested positive for coronavirus and No, I have not had coronavirus groups have significant difference in trust in politicians.

Conclusion

Firstly, while there was no significant association between vaccination status and signing petition ($p = 1$), vaccination status had a meaningful impact on political trust. The t-test showed a significant difference, with vaccinated individuals demonstrating higher trust in politicians. The effect size of 0.6 indicates a moderate to large impact, highlighting vaccination's role in shaping political trust. COVID-19 experience also significantly influences political trust. ANOVA showed that respondents who tested positive for coronavirus and respondents who have not had it have different levels of trust in politicians but the effect size is small so COVID-19 experience explains only a small portion of the variance in political trust. These results suggest that vaccination status and the direct experience of COVID-19 both affect political trust. Future research could explore other factors, such as demographic influences or government communication strategies, to better understand political trust and civic engagement.

References

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PROJECT 3

Topic

Public Perceptions and Responses to COVID-19 in Netherlands

Due to lack of data on our theme in 11 round of ESS, we have gone a bit off topic, but we still tried to add the idea of health to our analysis.

Research Question

To what extent do institutional trust, emotional attachment to the country, and subjective health shape overall life satisfaction among individuals in the Netherlands?

Relevance

According to the **World Happiness Report 2024**, the Netherlands ranks 6th among the happiest countries in the world (Helliwell et al., 2024). This high position suggests that people in the Netherlands are generally satisfied with their lives. Moreover, life satisfaction is an important part of well-being and is often used to understand how people feel about their quality of

life (Diener et al., 1999). Because of this, it is interesting and important to explore how emotional, institutional, and health-related aspects of life contribute to prosperity in a highly developed society.

Emotional attachment to the country is about how close or connected people feel to their country. Feeling proud or emotionally attached to one's country can help people feel more united and stable, which can make life feel better (Huddy & Khatib, 2007).

H1: *Higher emotional attachment to the country is positively associated with greater life satisfaction.*

Satisfaction with democracy is used to measure how much people trust political institutions. If people believe that the government is fair and works well, they may feel more secure and satisfied with their lives (Helliwell & Huang, 2008).

H2: *Greater satisfaction with democratic institutions is positively associated with greater life satisfaction.*

Satisfaction with health services is also important. When people are happy with the healthcare system, they may feel more supported by the state, which can improve their overall well-being (Alesina et al., 2004).

H3: *Greater satisfaction with the health care system is positively associated with greater life satisfaction.*

H4: *Emotional attachment to the country, satisfaction with democracy, and satisfaction with health services each significantly predict life satisfaction.*

We also include subjective general health because health is strongly related to how people feel about their lives. People who think their health is good are usually more satisfied, while those with poor health often feel less happy (Jylhä, 2009).

H5: *Subjective general health status significantly predicts life satisfaction, with better perceived health associated with higher life satisfaction.*

By focusing on these factors, this study helps us understand what shapes life satisfaction in the Netherlands. It connects personal feelings and health with views on how the country is doing, and shows how these things work together to influence well-being.

Data loading and selection

```
life_sat_ned <- read.csv('/Users/ludamuraveva/Desktop/DA/data/ESS11-subset.csv', sep = ',')

life_sat_ned_selected <- life_sat_ned %>%
  dplyr::select(atchctr, stfdem, stfhlth, stflife, health)

life_sat_ned_selected <- life_sat_ned_selected %>%
  mutate(
    atchctr = ifelse(atchctr %in% c(77,88,99), NA, atchctr),
    stfdem = ifelse(stfdem %in% c(77,88,99), NA, stfdem),
    stfhlth = ifelse(stfhlth %in% c(77,88,99), NA, stfhlth),
    stflife = ifelse(stflife %in% c(77,88,99), NA, stflife),
    health = ifelse(health %in% c(7,8,9), NA, health)
  ) %>%
  drop_na()
```

```
table(life_sat_ned_selected$atchctr)
```

```
##
##    0    1    2    3    4    5    6    7    8    9   10
##  32   15   36   50   50  116  196  351  483  197   90
```

Variables' Description

```
Variable_names <- c("atchctr", "stfdem", "health", "stfhlth","stflife")
Meanings <- c("How emotionally attached to Netherlands", "How satisfied with the way democracy works in country", "Subjective general health", "State of health services in country nowadays", "How satisfied with life as a whole")

Types <- c("Quasi-interval", "Quasi-interval", "Nominal", "Quasi-interval","Quasi-interval")

df <- data.frame(Variable_names, Meanings, Types, stringsAsFactors = FALSE)
kable(df) %>%
  kable_styling(bootstrap_options=c("bordered", "responsive","striped"), full_width = FALSE)
```

Variable_names	Meanings	Types
atchctr	How emotionally attached to Netherlands	Quasi-interval

Variable_names	Meanings	Types
stfdem	How satisfied with the way democracy works in country	Quasi-interval
health	Subjective general health	Nominal
stfhlth	State of health services in country nowadays	Quasi-interval
stflife	How satisfied with life as a whole	Quasi-interval

Data Types and Conversion

```
class(life_sat_ned_selected$atchctr)
```

```
## [1] "integer"
```

```
class(life_sat_ned_selected$stfdem)
```

```
## [1] "integer"
```

```
class(life_sat_ned_selected$stfhlth)
```

```
## [1] "integer"
```

```
class(life_sat_ned_selected$stflife)
```

```
## [1] "integer"
```

```
life_sat_ned_selected$health <- factor(life_sat_ned_selected$health,
                                       levels = c(1, 2, 3, 4, 5),
                                       labels = c("Very good", "Good", "Fair", "Bad", "Very bad"))
```

Descriptive statistics

```
vars <- life_sat_ned_selected[, c("atchctr", "stfdem", "stfhlth", "stflife")]
```

```
describe(vars)
```

```
##          vars      n mean  sd median trimmed  mad min max range  skew kurtosis
## atchctr    1 1616 6.92 2.08      7    7.19 1.48   0  10   10 -1.26    1.71
## stfdem     2 1616 5.98 2.08      6    6.18 1.48   0  10   10 -0.81    0.40
## stfhlth    3 1616 6.61 1.78      7    6.76 1.48   0  10   10 -0.93    0.99
## stflife    4 1616 7.90 1.39      8    8.06 1.48   0  10   10 -1.93    7.44
##              se
## atchctr 0.05
## stfdem  0.05
## stfhlth 0.04
## stflife 0.03
```

```
get_mode <- function(x) {
  ux <- unique(x)
  ux[which.max(tabulate(match(x, ux)))]
}
```

```
sapply(life_sat_ned_selected[, c("atchctr", "stfdem", "stfhlth", "stflife", 'health')], get_mode)
```

```
## atchctr  stfdem stfhlth stflife  health
##        8       7       8       8      2
```

Although mean and median everywhere are close to each other, there is negative skewness and light or heavy tails in some cases.

atchctr - slightly left-skewed and lighter tails

stfdem - light tails

stfhlth - light tails

stflife - highly left-skewed and heavy tails

Modes suggest that the most common responses for satisfaction and attachment items are relatively high (7/8 out of 10), indicating overall positive evaluations in these domains, which are quite close to mean and medians.

The mode for self-rated health is also pretty high (2), which means 'Good' in codebook.

Boxplots

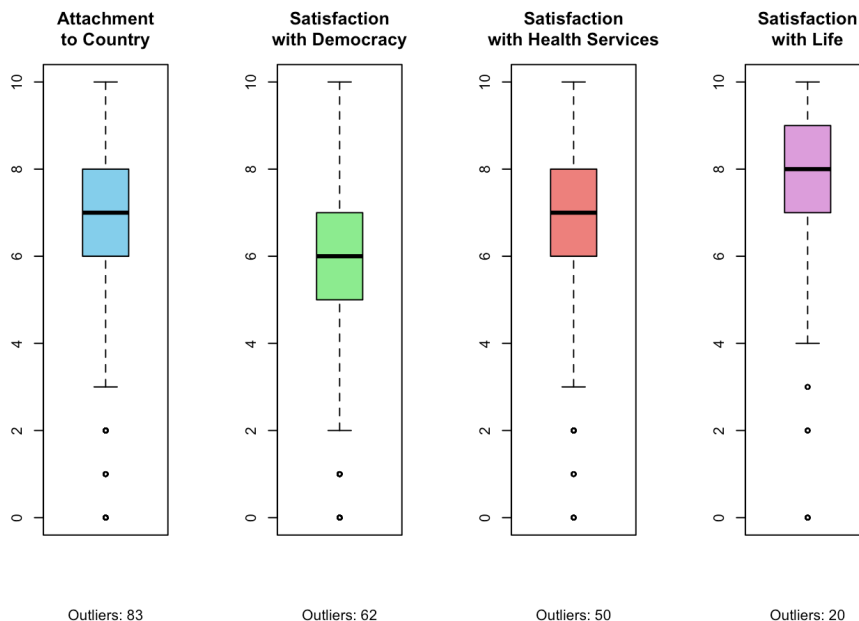
```
par(mfrow = c(1, 4),
    mar = c(5, 4, 4, 2) + 0.1,
    oma = c(0, 0, 2, 0))

boxplot(life_sat_ned_selected$atchctr,
        main = "Attachment\nto Country",
        sub = paste("Outliers:", length(boxplot.stats(life_sat_ned_selected$atchctr)$out)),
        col = "skyblue")

boxplot(life_sat_ned_selected$stfdem,
        main = "Satisfaction\nwith Democracy",
        sub = paste("Outliers:", length(boxplot.stats(life_sat_ned_selected$stfdem)$out)),
        col = "lightgreen")

boxplot(life_sat_ned_selected$stfhlth,
        main = "Satisfaction\nwith Health Services",
        sub = paste("Outliers:", length(boxplot.stats(life_sat_ned_selected$stfhlth)$out)),
        col = "lightcoral")

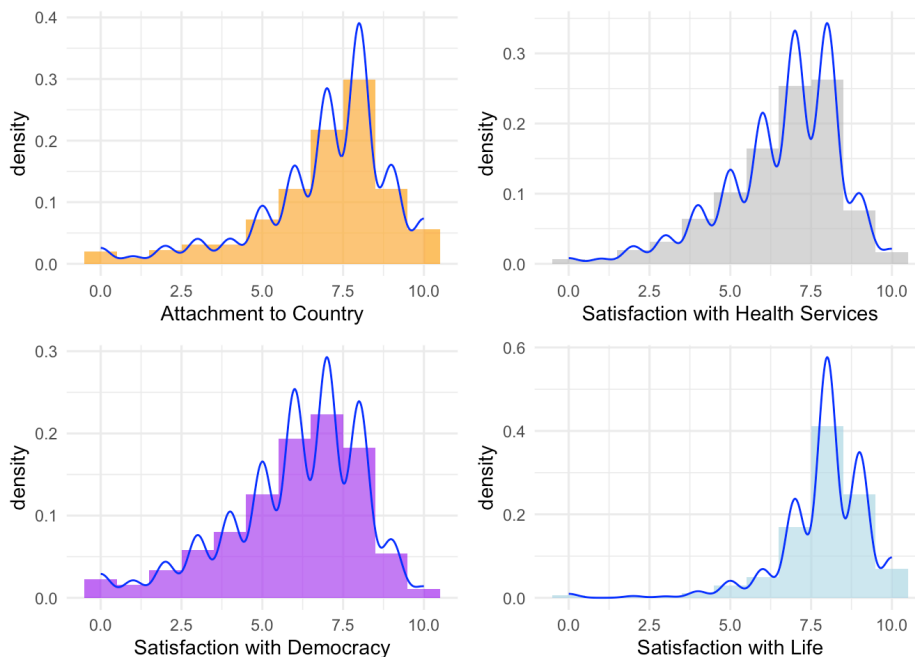
boxplot(life_sat_ned_selected$stflife,
        main = "Satisfaction\nwith Life",
        sub = paste("Outliers:", length(boxplot.stats(life_sat_ned_selected$stflife)$out)),
        col = "plum")
```



Comment: As we can see all variables have outliers. Medians are from 6 to 8. The most symmetrical is satisfaction with democracy (also having biggest spread), others are more left-skewed.

Histograms

```
emo <- ggplot(data = life_sat_ned_selected, aes(x = atchctr)) +  
  geom_histogram(aes(y = after_stat(density)), position = "identity", alpha = 0.7, binwidth = 1, fill = "orange") +  
  geom_density(col = "blue", fill = "white", alpha = 0.1) +  
  xlab("Attachment to Country") +  
  theme_minimal()  
  
dem <- ggplot(data = life_sat_ned_selected, aes(x = stfdem)) +  
  geom_histogram(aes(y = after_stat(density)), position = "identity", alpha = 0.7, binwidth = 1, fill = "purple") +  
  geom_density(col = "blue", fill = "white", alpha = 0.1) +  
  xlab("Satisfaction with Democracy") +  
  theme_minimal()  
  
health <- ggplot(data = life_sat_ned_selected, aes(x = stfhlth)) +  
  geom_histogram(aes(y = after_stat(density)), position = "identity", alpha = 0.7, binwidth = 1, fill = "grey") +  
  geom_density(col = "blue", fill = "white", alpha = 0.1) +  
  xlab("Satisfaction with Health Services") +  
  theme_minimal()  
  
life <- ggplot(data = life_sat_ned_selected, aes(x = stflife)) +  
  geom_histogram(aes(y = after_stat(density)), position = "identity", alpha = 0.7, binwidth = 1, fill = "lightblue") +  
  geom_density(col = "blue", fill = "white", alpha = 0.1) +  
  xlab("Satisfaction with Life") +  
  theme_minimal()  
  
plot_grid(emo, health, dem, life, ncol = 2)
```

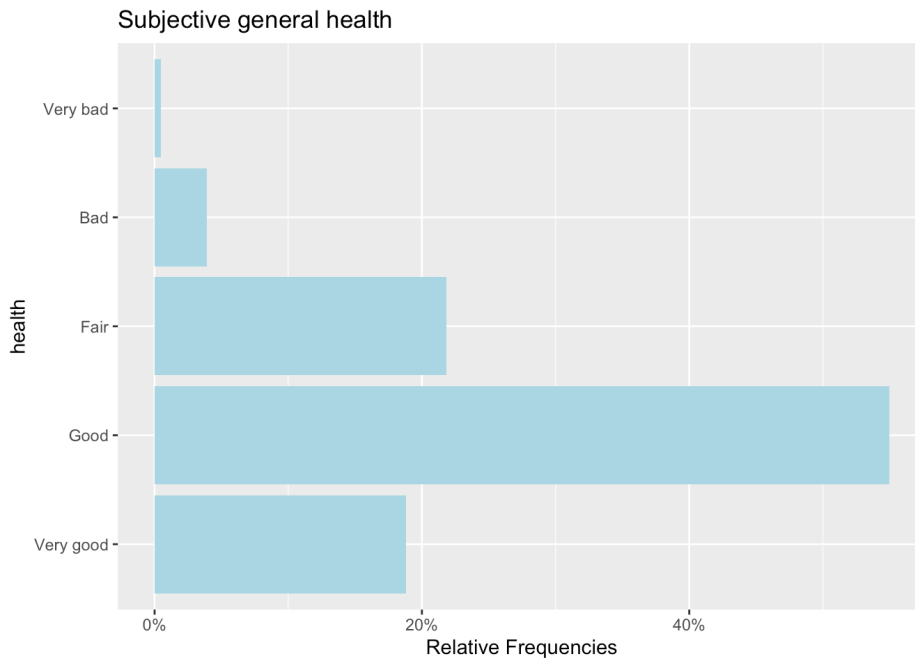


Comment: According to plots, all variables are left-skewed -> not normally distributed.

- 1 - Attachment to country has low spread of answers with mostly above 5.
- 2 - Satisfaction with health services seems having bimodal distribution with a bit more spread of answers.
- 3 - Satisfaction with democracy has more symmetrical distribution and wider range of responses on scale.
- 4 - Satisfaction with life is the most peaked one, most responses are concentrated at the top.

Barplot

```
ggplot(data = life_sat_ned_selected, aes(x = health)) +  
  geom_bar(aes(y = after_stat(count) / sum(after_stat(count))), fill = "lightblue") +  
  scale_y_continuous(labels = scales::percent) +  
  ylab("Relative Frequencies") +  
  ggtitle("Subjective general health") +  
  coord_flip()
```



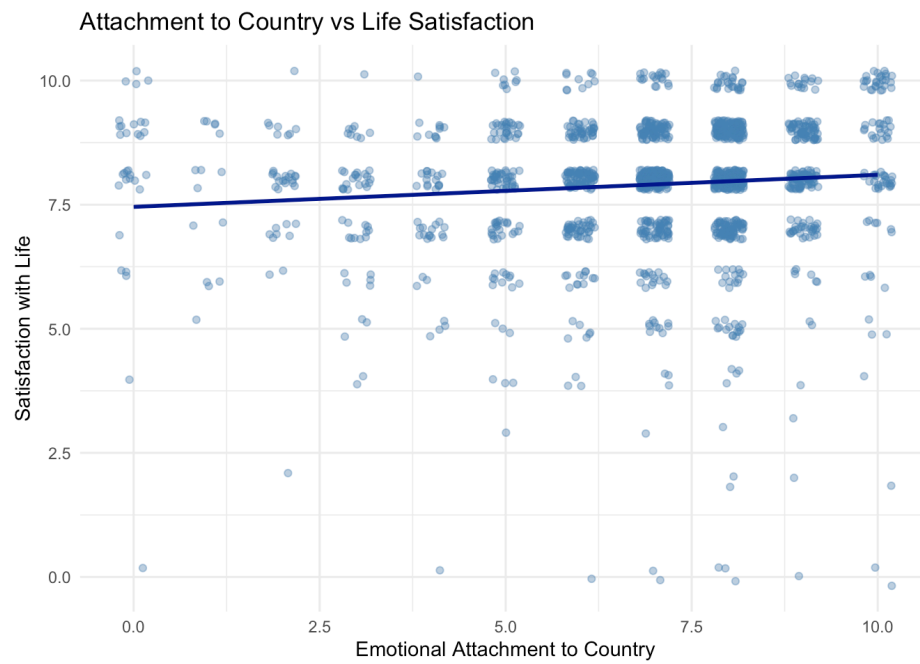
Comment: As the category 'Very bad' have very few observations, we join it with other category 'Bad' as they are close by their meaning. According to plot, most people consider themselves having good health.

Join of categories

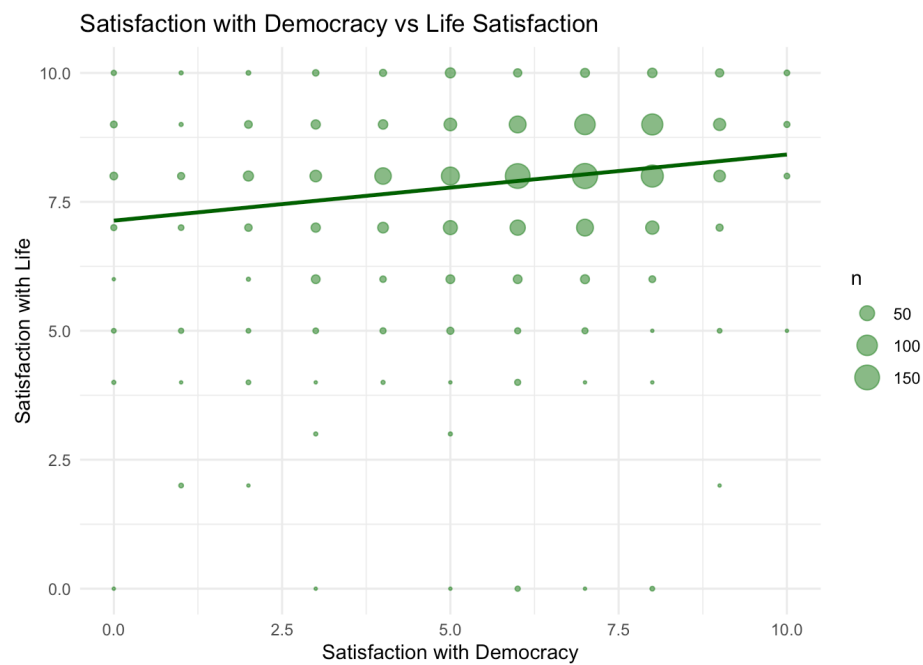
```
life_sat_ned_selected$health_collapsed <- fct_collapse(  
  life_sat_ned_selected$health,  
  "Very good" = "Very good",  
  "Good" = "Good",  
  "Fair" = "Fair",  
  "Poor" = c("Bad", "Very bad")  
)  
# Join of categories with few frequencies
```

Scatterplots

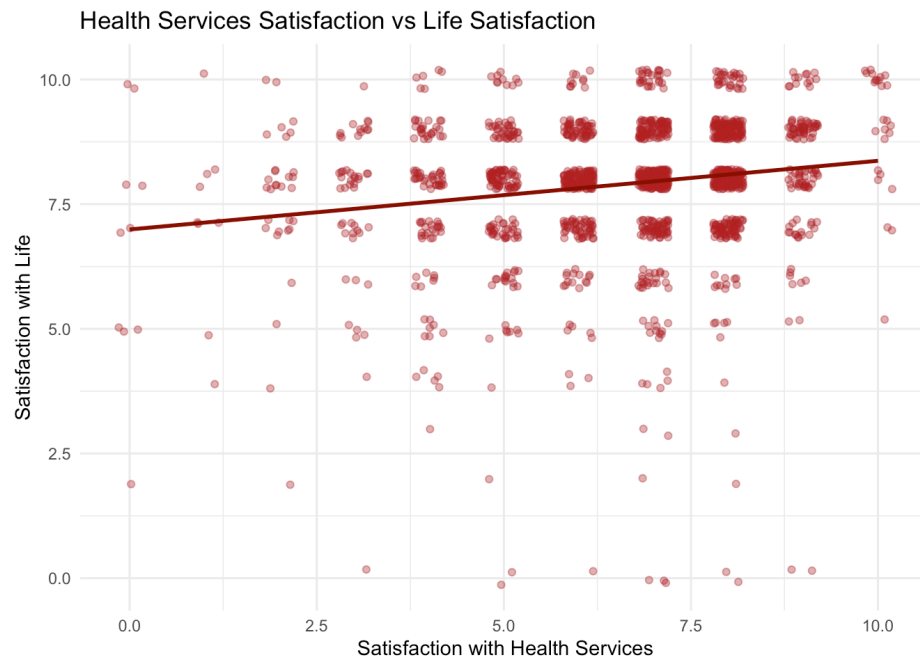
```
ggplot(life_sat_ned_selected, aes(x = atchctr, y = stflife)) +  
  geom_jitter(width = 0.2, height = 0.2, alpha = 0.4, color = "steelblue") +  
  geom_smooth(method = "lm", se = FALSE, color = "darkblue") +  
  labs(  
    title = "Attachment to Country vs Life Satisfaction",  
    x = "Emotional Attachment to Country",  
    y = "Satisfaction with Life"  
  ) +  
  theme_minimal()
```



```
ggplot(life_sat_ned_selected, aes(x = stfdem, y = stflife)) +
  geom_count(aes(size = ..n..), alpha = 0.6, color = "forestgreen") +
  scale_size_area(max_size = 6) +
  geom_smooth(method = "lm", se = FALSE, color = "darkgreen") +
  labs(
    title = "Satisfaction with Democracy vs Life Satisfaction",
    x = "Satisfaction with Democracy",
    y = "Satisfaction with Life"
  ) +
  theme_minimal()
```



```
ggplot(life_sat_ned_selected, aes(x = stfhlth, y = stflife)) +
  geom_jitter(width = 0.2, height = 0.2, alpha = 0.4, color = "firebrick") +
  geom_smooth(method = "lm", se = FALSE, color = "darkred") +
  labs(
    title = "Health Services Satisfaction vs Life Satisfaction",
    x = "Satisfaction with Health Services",
    y = "Satisfaction with Life"
  ) +
  theme_minimal()
```



Comment: The upward lines on 3 plots indicate positive relationships of variables with life satisfaction. However, dots are not nearly clustered around the line. That is why, we cannot tell about strong correlations. In plots we can observe heteroscedasticity, as dots fan out like a cone. The predictors might be related to life satisfaction in similar ways. Moreover, life satisfaction might be more influenced by other factors like income, etc.

Correlation matrix (Spearman)

```
life_sat_cont <- life_sat_ned_selected %>% dplyr::select(atchctr, stfdem, stfhlth, stflife)

cor_matrix <- cor(life_sat_cont, method = "spearman")
```

```
tab_corr(cor_matrix,
  title = "Spearman Correlation Matrix",
  show.p = TRUE,
  digits = 2)
```

Spearman Correlation Matrix

	<i>atchctr</i>	<i>stfdem</i>	<i>stfhlth</i>	<i>stflife</i>
<i>atchctr</i>		0.17	0.11	0.14
<i>stfdem</i>	0.17		0.36	0.20
<i>stfhlth</i>	0.11	0.36		0.18
<i>stflife</i>	0.14	0.20	0.18	

Computed correlation used pearson-method with listwise-deletion.

Comment: Due to non-normal distribution we used Spearman for measuring correlation, which is more robust to such violations. According to correlation coefficients, all three variables are weakly correlate with life satisfaction. Also as we supposed some of the predictors correlate with each other (Satisfaction with Health Services & Satisfaction with Democracy) = 0.37, which is moderate correlation. Other variables have weak correlation.

P-values

```
corr_results <- corr.test(life_sat_cont, method = "spearman")
corr_results$p
```

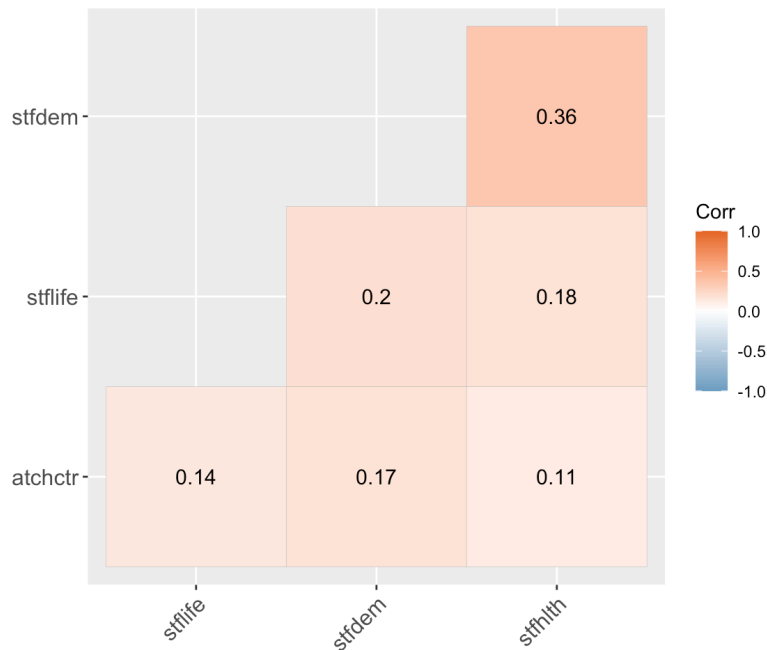
```
##          atchctr      stfdem      stfhlth      stflife
## atchctr 0.000000e+00 2.135549e-11 8.609374e-06 2.689959e-08
## stfdem  7.118497e-12 0.000000e+00 2.565771e-51 2.188273e-15
## stfhlth 8.609374e-06 4.276284e-52 0.000000e+00 3.302047e-13
## stflife 1.344980e-08 4.376545e-16 8.255118e-14 0.000000e+00
```

Comment: All $p < 0.001$, meaning the results of Spearman correlation are highly significant.

Correlation Heatmap

```
p_values <- cor_pmat(life_sat_cont)

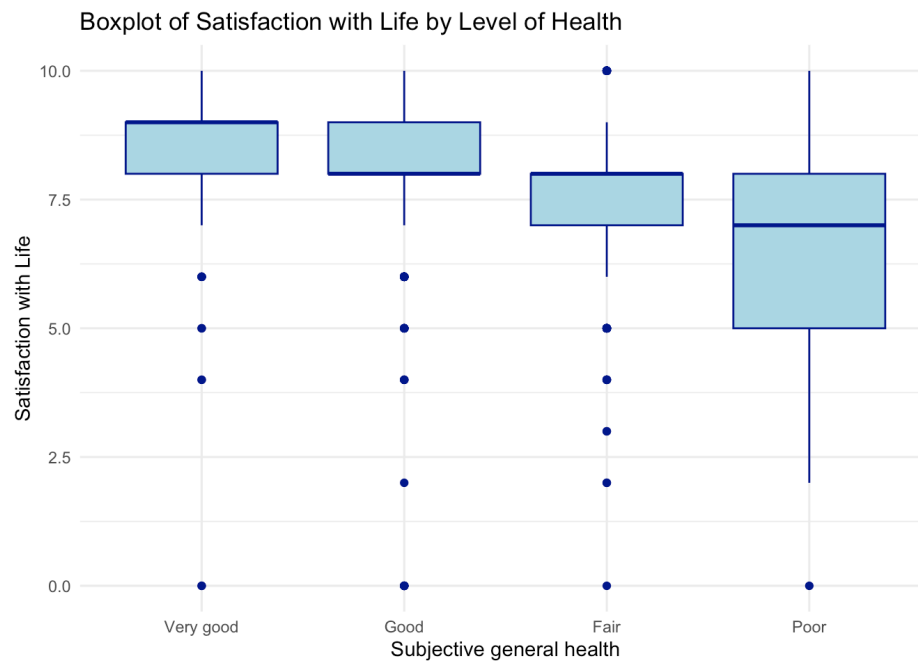
ggcorrplot(cor_matrix,
            hc.order = TRUE,
            type = "lower",
            p.mat = p_values,
            insig = "blank",
            lab = TRUE,
            ggtheme = ggplot2::theme_gray(),
            colors = c("#6D9EC1", "white", "#E46726"))
```



Comment: Here is the visual representation of correlation's strength.

Boxplot for the categorical predictor

```
ggplot(data = life_sat_ned_selected, aes(x = health_collapsed, y = stflife)) +
  geom_boxplot(fill = "lightblue", color = "darkblue") +
  xlab("Subjective general health") +
  ylab("Satisfaction with Life") +
  ggtitle("Boxplot of Satisfaction with Life by Level of Health") +
  theme_minimal()
```



Comment: Here we can observe, how with increasing in health perception, satisfaction with life increases accordingly. Also the health category 'Poor' shows the biggest variability in responses, while better health groups show narrow spread, clustering near the top of the scale. In addition, all of them contain outliers.

Building regression model

For the regression model we decided to use same three continuous predictors:

- 1) **atchctr** (How emotionally attached to Netherlands)
- 2) **stfdem** (How satisfied with the way democracy works in country)
- 3) **stfhlt** (State of health services in country nowadays)

one categorical predictor: **health_collapsed** (Subjective general health)

outcome variable: **stflife** (Satisfaction with life as a whole)

```
m1 <- lm(stflife ~ atchctr, data = life_sat_ned_selected)
sjPlot::tab_model(m1)
```

stflife			
Predictors	Estimates	CI	p
(Intercept)	7.46	7.22 – 7.69	<0.001
atchctr	0.06	0.03 – 0.10	<0.001
Observations	1616		
R ² / R ² adjusted	0.009 / 0.009		

The model shows a small but statistically significant positive association between attachment to country and life satisfaction.

R² = 0.009 indicates that this model explains less than 1% of the variance in life satisfaction.

```
m2 <- lm(stflife ~ atchctr + stfdem, data = life_sat_ned_selected)
sjPlot::tab_model(m2)
```

stflife			
Predictors	Estimates	CI	p
(Intercept)	6.89	6.62 – 7.17	<0.001
atchctr	0.04	0.01 – 0.07	0.012
stfdem	0.12	0.09 – 0.15	<0.001
Observations	1616		
R ² / R ² adjusted	0.041 / 0.039		

Adding satisfaction with democracy significantly improves the model. Both predictors are significant, with satisfaction with democracy showing a stronger effect. R^2 increases to 0.041, suggesting that the model now explains 4.1% of the variance—an improvement of over 3% compared to Model 1.

```
m3 <- lm(stflife ~ atchctr + stfdem + stfhlth, data = life_sat_ned_selected)
sjPlot::tab_model(m3)
```

stflife			
Predictors	Estimates	CI	p
(Intercept)	6.49	6.16 – 6.82	<0.001
atchctr	0.04	0.01 – 0.07	0.016
stfdem	0.09	0.05 – 0.12	<0.001
stfhlth	0.09	0.05 – 0.13	<0.001
Observations	1616		

R^2 / R^2 adjusted 0.052 / 0.050

Including satisfaction with health services further improves the model. All predictors remain significant, and the new predictor also positively contributes.

R^2 increases to 0.052, which shows a modest improvement, adding about 1% more explained variance.

```
m4 <- lm(stflife ~ atchctr + stfdem + stfhlth + health_collapsed, data = life_sat_ned_selected)
sjPlot::tab_model(m4)
```

stflife			
Predictors	Estimates	CI	p
(Intercept)	7.24	6.88 – 7.60	<0.001
atchctr	0.04	0.01 – 0.07	0.010
stfdem	0.06	0.03 – 0.09	0.001
stfhlth	0.08	0.04 – 0.11	<0.001
health collapsed [Good]	-0.41	-0.58 – -0.24	<0.001
health collapsed [Fair]	-0.82	-1.03 – -0.62	<0.001
health collapsed [Poor]	-1.64	-1.98 – -1.29	<0.001
Observations	1616		

R^2 / R^2 adjusted 0.117 / 0.113

Health status is a strong predictor. As health declines, life satisfaction drops significantly.

R^2 increases to 0.117, showing a large improvement compared to previous models.

This final model explains nearly 12% of the variance in life satisfaction.

With each step, the R^2 value increases, indicating that the added predictors significantly improve the model's ability to explain life satisfaction.

Comparing Models

```
anova(m1, m2)
```

```
## Analysis of Variance Table
##
## Model 1: stflife ~ atchctr
## Model 2: stflife ~ atchctr + stfdem
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1   1614 3085.5
## 2   1613 2988.3   1    97.177 52.453 6.795e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m2,m3)
```

```
## Analysis of Variance Table
##
## Model 1: stflife ~ atchctr + stfdem
## Model 2: stflife ~ atchctr + stfdem + stfhlth
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     1613 2988.3
## 2     1612 2951.9   1     36.371 19.861 8.901e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m3,m4)
```

```
## Analysis of Variance Table
##
## Model 1: stflife ~ atchctr + stfdem + stfhlth
## Model 2: stflife ~ atchctr + stfdem + stfhlth + health_collapsed
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     1612 2951.9
## 2     1609 2750.8   3     201.14 39.216 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

According to comparison of our models, each addition of a predictor significantly improves the model ($p < 0.001$), reducing the residual variance and increasing the explained variation in life satisfaction. Subjective health (health_collapsed) makes the largest contribution to the improvement of the model, as adding this variable reduces RSS by 201.14 and has the highest F-statistic. Thus, all predictors - emotional attachment, satisfaction with democracy, satisfaction with health care, and subjective health - are significantly associated with life satisfaction and together explain its variation better than any of the models with fewer variables. So we choose model 4 for our further analysis.

Interpretation of coefficients

```
sjPlot::tab_model(m4)
```

Predictors	stflife		
	Estimates	CI	p
(Intercept)	7.24	6.88 – 7.60	<0.001
atchctr	0.04	0.01 – 0.07	0.010
stfdem	0.06	0.03 – 0.09	0.001
stfhlth	0.08	0.04 – 0.11	<0.001
health collapsed [Good]	-0.41	-0.58 – -0.24	<0.001
health collapsed [Fair]	-0.82	-1.03 – -0.62	<0.001
health collapsed [Poor]	-1.64	-1.98 – -1.29	<0.001
Observations	1616		
R ² / R ² adjusted	0.117 / 0.113		

Comment:

- The intercept represents that expected life satisfaction for a person with “Very good” health category is 7.24, while other predictors are 0.
- With each 1 unit increase in attachment to country, life satisfaction increases by 0.04, controlling for other predictors. The effect is positive but small. It means people who are more emotionally attached to Netherlands tend to have higher life satisfaction.
- With each 1 unit increase in satisfaction with democracy, life satisfaction increases by 0.06, controlling for other predictors. It means people who are more satisfied with democracy tend to have higher life satisfaction. The effect is also small and positive.
- With each 1 unit increase in satisfaction with health services, life satisfaction increases by 0.08, holding other predictors constant. The effect is positive but small. It suggests people who are more satisfied with health services tend to have higher life satisfaction.

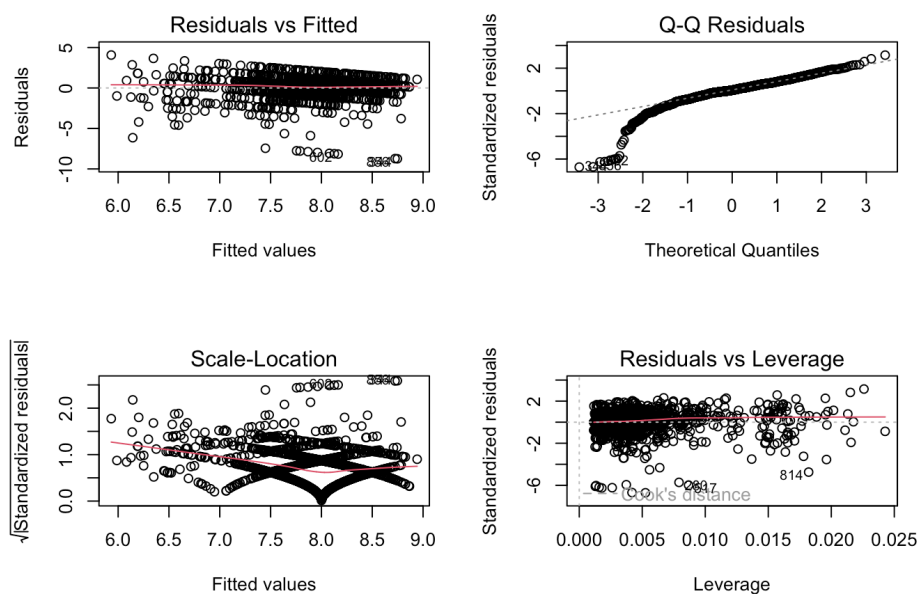
- People with good health have, on average, 0.41 units lower life satisfaction than those with very good health, holding other variables constant.
- People with fair health have, on average, 0.82 units lower life satisfaction than those with very good health.
- People with poor health have, on average, 1.64 units lower life satisfaction compared to those with very good health.
- As expected, lower health categories are associated with a significant decline in life satisfaction.
- Adjusted $R^2 = 0.113$ means that model explains 11.3% of the variance in life satisfaction.
- All predictors in our model have a statistically significant relationship with life satisfaction.

Regression Equation

$stflife = 7.24 + 0.04 \times atchctr + 0.06 \times stfdem + 0.08 \times stfhlth - 0.41 \times health_Good - 0.82 \times health_Fair - 1.64 \times health_Poor$

Linear regression assumptions

```
par(mfrow = c(2,2))
plot(m4)
```



```
# for homoscedasticity
bptest(m4)
```

```
##
## studentized Breusch-Pagan test
##
## data: m4
## BP = 23.971, df = 6, p-value = 0.0005286
```

1 - In the Residuals vs Fitted plot, the residuals appear randomly distributed along the horizontal line at zero, which supports the assumption of linearity between predictors and the response.

2 - In the Q-Q plot, most residuals closely follow the diagonal line, suggesting approximate normality. However, the presence of heavy tails indicates some deviations from a perfect normal distribution.

3 - In the Scale-Location plot, the red line is fairly horizontal, indicating constant variance (homoscedasticity), although there is a slight decline toward the right end, which may suggest increasing or decreasing variance. To confirm, I conducted a formal test (Breusch-Pagan), and since $p < 0.05$, we reject the null hypothesis of homoscedasticity, indicating that heteroscedasticity may be present.

4 - As for Residuals vs Leverage plot we see 3 labeled points below red line which is threshold for influential outliers, which may pull the regression.

Conclusion

This study aimed to investigate how emotional attachment to the country, satisfaction with democracy, satisfaction with health services, and subjective general health are associated with life satisfaction in the Netherlands. Our findings provide partial support for the hypotheses.*

Our analysis showed that higher emotional attachment to the country is significantly associated with greater life satisfaction, supporting Hypothesis 1. This suggests that individuals who feel more emotionally connected to their country generally report higher life satisfaction.

Similarly, greater satisfaction with democratic institutions was significantly associated with life satisfaction, in line with Hypothesis 2. This means that individuals who view the democratic system positively tend to have higher life satisfaction.

Regarding Hypothesis 3, satisfaction with health services was also a significant predictor of life satisfaction. This indicates that individuals who are more satisfied with the healthcare system tend to feel more satisfied with their overall lives.

Subjective general health (Hypothesis 5) was found to be a strong predictor of life satisfaction. People who perceive their health as better report higher life satisfaction, highlighting the critical role of health in well-being.

Hypothesis 4, which suggested that these factors would significantly predict life satisfaction together, was confirmed. However, it is important to note that the model explained only 11.3% of the variance in life satisfaction. Although this percentage is relatively small, it is still meaningful in social research, where human behavior is influenced by many factors. Moreover, it is worth considering that the Netherlands ranks among the top 10 countries globally by GDP per capita, a factor that likely contributes to the overall high life satisfaction levels in the country (World Bank, 2024). The high standard of living, access to social services, and economic prosperity in the Netherlands may serve as a context in which these predictors are assessed, potentially influencing life satisfaction beyond the immediate factors captured in the model.

These results emphasize that emotional, institutional, and health-related factors are all relevant to understanding life satisfaction in the Netherlands.

*Although some assumptions were violated, the model still has value for understanding relationships between predictors and the outcome, but statistical significance and confidence intervals might be unreliable.

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PROJECT 4

Theoretical Background

Subjective well-being, often operationalized as **life satisfaction**, is influenced by a complex interplay of individual, social, and institutional factors. In this study, we investigate the roles of health, social interaction, and institutional trust in shaping life satisfaction among individuals in the Netherlands. The theoretical foundation of this project draws primarily from the **Social Determinants of Health model** and **Social Capital Theory**, which together emphasize both structural and relational factors as key influences on well-being.

According to the **SDH framework**, health outcomes—and by extension, subjective well-being—are not merely the result of individual behaviors or genetic predispositions, but are shaped by the broader social, economic, and political environments in which people live (World Health Organization [WHO], 2008). Key among these determinants is **individual health status**, which

has consistently been shown to be a strong predictor of life satisfaction (Diener et al., 2018; Helliwell et al., 2020). People who perceive themselves to be in good health typically report higher levels of well-being, both because of the intrinsic value of health and because of the way health enables participation in various aspects of social and economic life (Subramanian et al., 2005).

Social Capital Theory further highlights the importance of social relationships and networks in supporting well-being. Social capital, broadly defined as the resources available to individuals through their social connections, has been positively linked to both mental and physical health, as well as overall life satisfaction (Putnam, 2000; Kawachi & Berkman, 2001). Frequent social interaction, particularly face-to-face meetings with friends and family, has been found to foster emotional support, a sense of belonging, and resilience in the face of stress (Helliwell & Putnam, 2004). Thus, we hypothesize that the **frequency of social contact** will be positively associated with life satisfaction.

Moreover, the interaction between health and social contact may be particularly important. Prior research suggests that social engagement can amplify the benefits of good health for subjective well-being, as people in better health may be more capable of engaging in and enjoying social activities (Kawachi & Berkman, 2001). We therefore explore whether the association between health and life satisfaction is moderated by the frequency of social interaction.

Finally, trust in institutions, such as democratic systems and healthcare services, constitutes another crucial dimension of subjective well-being. **Institutional trust** reflects individuals' confidence that public institutions are competent, fair, and responsive, which can reduce feelings of uncertainty and insecurity in society (Rothstein & Stolle, 2008). In the context of **the Netherlands—a high-income democracy with a well-established welfare**—trust in democratic institutions and satisfaction with healthcare provision are expected to contribute positively to life satisfaction by reinforcing a sense of social stability and procedural justice (Helliwell et al., 2021).

Taken together, these perspectives suggest that life satisfaction is driven by a combination of individual-level factors (e.g., health), relational resources (e.g., social interaction), and macro-level perceptions (e.g., institutional trust). Our analysis investigates these relationships within a Dutch context, while also considering possible interaction effects between key variables.

Research Questions

1. To what extent do subjective health, frequency of social contact, trust in democratic institutions, and satisfaction with the healthcare system independently predict life satisfaction among individuals in the Netherlands?

2. Does the frequency of social contact moderate the relationship between subjective health and life satisfaction, after controlling for trust in democratic institutions and satisfaction with the healthcare system?

Hypotheses

H1: *Individuals who report better subjective health will also report higher levels of life satisfaction.*

H2: *Individuals who engage in more frequent face-to-face social interactions will report higher levels of life satisfaction.*

H3: *The positive effect of social interaction on life satisfaction will be stronger among individuals with poorer subjective health than among those in good health.*

H4: *Individuals with higher trust in democratic institutions will report higher levels of life satisfaction.*

H5: *Individuals who express greater satisfaction with the healthcare system will report higher levels of life satisfaction.*

H6: *The positive relationship between subjective health and life satisfaction depends on the frequency of social contact, such that the effect of good health on life satisfaction is stronger among those who socialize more frequently.*

Data loading and selection

```
life_sat_ned <- read.csv('/Users/ludamuraveva/Desktop/DA/data/ESS11-subset.csv', sep = ',')

life_sat_ned_selected <- life_sat_ned %>%
  dplyr::select(stfdem, stfhlth, stflife, health, sclmeet)

life_sat_ned_selected<- life_sat_ned_selected %>%
  mutate(
    stfdem = ifelse(stfdem %in% c(77,88,99), NA, stfdem),
    stfhlth = ifelse(stfhlth %in% c(77,88,99), NA, stfhlth),
    stflife = ifelse(stflife %in% c(77,88,99), NA, stflife),
    health = ifelse(health %in% c(7,8,9), NA, health),
    sclmeet = ifelse(sclmeet %in% c(77,88,99), NA, sclmeet)
  ) %>%
  drop_na()
```

Variables' Description

```
Variable_names <- c("stfdem", "health", "stfhlth", "stflife", "sclmeet")
Meanings <- c("How satisfied with the way democracy works in country", "Subjective general health",
"State of health services in country nowadays", "How satisfied with life as a whole", "How often so
cially meet with friends, relatives or colleagues")

Types <- c("Quasi-interval", "Nominal", "Quasi-interval", "Quasi-interval", "Nominal")

df <- data.frame(Variable_names, Meanings, Types, stringsAsFactors = FALSE)
kable(df) %>%
  kable_styling(bootstrap_options=c("bordered", "responsive", "striped"), full_width = FALSE)
```

Variable_names	Meanings	Types
stfdem	How satisfied with the way democracy works in country	Quasi-interval
health	Subjective general health	Nominal
stfhlth	State of health services in country nowadays	Quasi-interval
stflife	How satisfied with life as a whole	Quasi-interval
sclmeet	How often socially meet with friends, relatives or colleagues	Nominal

Subjective Health → Life Satisfaction Individuals with better subjective health tend to report higher life satisfaction. This link is well established in research, as good health enables individuals to participate fully in daily life, maintain independence, and avoid chronic stress or disability (Diener & Chan, 2011). Subjective health reflects not only physical well-being but also a sense of vitality and psychological control, both of which are central to well-being.

Social Contact → Life Satisfaction Frequent social interaction supports life satisfaction by fulfilling emotional and psychological needs. Meaningful contact with others provides companionship, emotional support, and a buffer against stress (Cohen, 2004). It also reinforces a sense of belonging and social identity. People who socialize more often typically report greater happiness and lower loneliness, which directly contribute to their overall life satisfaction.

Satisfaction with Democracy → Life Satisfaction Trust in democratic institutions is positively linked to life satisfaction because it reflects confidence in political fairness, citizen voice, and governance legitimacy. When individuals believe their society operates on principles of justice and representation, they are more likely to feel secure, respected, and hopeful about the future (Helliwell & Huang, 2008). Satisfaction with democracy can also enhance a sense of autonomy and societal belonging.

Satisfaction with Healthcare → Life Satisfaction Belief in the reliability and quality of the healthcare system is another key contributor to life satisfaction. People who trust the healthcare system tend to feel less anxious about illness and more confident in their ability to access care when needed. This trust reduces uncertainty around health issues and contributes to a stronger sense of safety and overall well-being (Helliwell & Huang, 2008).

Data Types and Conversion

```
class(life_sat_ned_selected$stfdem)
```

```
## [1] "integer"
```

```
class(life_sat_ned_selected$stfhlth)
```

```
## [1] "integer"
```

```
class(life_sat_ned_selected$stflife)
```

```
## [1] "integer"
```

```
class(life_sat_ned_selected$sclmeet)
```

```
## [1] "integer"
```

```

life_sat_ned_selected$sclmeet <- factor(life_sat_ned_selected$sclmeet)

life_sat_ned_selected$health <- factor(life_sat_ned_selected$health,
                                       levels = c(1, 2, 3, 4, 5),
                                       labels = c("Very good", "Good", "Fair", "Bad", "Very bad"))

life_sat_ned_selected$health_collapsed <- fct_collapse(
  life_sat_ned_selected$health,
  "Very good" = "Very good",
  "Good" = "Good",
  "Fair" = "Fair",
  "Poor" = c("Bad", "Very bad")
)

life_sat_ned_selected$sclmeet <- factor(life_sat_ned_selected$sclmeet,
                                       levels = c(1, 2, 3, 4, 5, 6, 7),
                                       labels = c("Never",
                                                  "Less than once a month",
                                                  "Once a month",
                                                  "Several times a month",
                                                  "Once a week",
                                                  "Several times a week",
                                                  "Every day"))

life_sat_ned_selected$sclmeet_collapsed <- fct_collapse(
  life_sat_ned_selected$sclmeet,
  "Once a month or less" = c("Never", "Less than once a month", "Once a month"),
  "Several times a month" = "Several times a month",
  "Once a week" = "Once a week",
  "Several times a week" = "Several times a week",
  "Every day" = "Every day"
)

```

Descriptive Statistics

```
summary(life_sat_ned_selected)
```

```

##      stfdem      stfhlth      stflife      health
## Min.   : 0.000   Min.   : 0.000   Min.   : 0.000   Very good:304
## 1st Qu.: 5.000   1st Qu.: 6.000   1st Qu.: 7.000   Good    :888
## Median : 6.000   Median : 7.000   Median : 8.000   Fair    :354
## Mean   : 5.981   Mean   : 6.609   Mean   : 7.901   Bad     : 64
## 3rd Qu.: 7.000   3rd Qu.: 8.000   3rd Qu.: 9.000   Very bad: 8
## Max.   :10.000   Max.   :10.000   Max.   :10.000
##
##              sclmeet   health_collapsed   sclmeet_collapsed
## Never           : 8   Very good:304   Once a month or less :123
## Less than once a month: 40   Good    :888   Several times a month:286
## Once a month      : 75   Fair    :354   Once a week          :241
## Several times a month:286   Poor    : 72   Several times a week :702
## Once a week        :241                      Every day            :266
## Several times a week :702
## Every day          :266

```

```
table(life_sat_ned_selected$health_collapsed)
```

```

##
## Very good   Good   Fair   Poor
##      304     888     354     72

```

```
table(life_sat_ned_selected$sclmeet_collapsed)
```

```
##
## Once a month or less Several times a month Once a week
## 123 286 241
## Several times a week Every day
## 702 266
```

The means and medians for the satisfaction variables (**stfdem**, **stfhlth**, **stflife**) are relatively close, suggesting that their distributions are approximately symmetric, though not perfectly normal. Specifically:

Trust in democracy (**stfdem**) and trust in the health system (**stfhlth**) show mild right skewness, with most responses concentrated between 5 and 7.

Life satisfaction (**stflife**) appears left-skewed, with a clustering of high values around 8–9, but a noticeable tail of lower responses, indicating that while most respondents report high life satisfaction, a minority report low satisfaction.

The most frequent scores (**modes**) for the **satisfaction items** fall between 6 and 8, aligning with their central tendencies and reinforcing a generally positive outlook among respondents.

For **self-rated health**, the most common category is “**Good**” followed by “Fair” and “Very good,” with relatively few respondents rating their health as “Poor”.

In terms of **social contact**, a large portion of respondents report meeting socially **several times a week**, indicating frequent interpersonal interaction. Fewer respondents report rare social contact (once a month or less), though this group still constitutes a noticeable minority. This variation highlights potential differences in social integration within the population.

*Visualizations can be found in Project 3.

Additive regression model

```
additive_model <- lm(stflife ~ health_collapsed + sclmeet_collapsed + stfdem + stfhlth, data = life_sat_ned_selected)

sjPlot::tab_model(additive_model)
```

Predictors	stflife		
	Estimates	CI	p
(Intercept)	7.14	6.76 – 7.52	<0.001
health collapsed [Good]	-0.38	-0.55 – -0.21	<0.001
health collapsed [Fair]	-0.79	-1.00 – -0.59	<0.001
health collapsed [Poor]	-1.54	-1.88 – -1.19	<0.001
sclmeet collapsed [Several times a month]	0.25	-0.03 – 0.53	0.078
sclmeet collapsed [Once a week]	0.23	-0.06 – 0.51	0.116
sclmeet collapsed [Several times a week]	0.41	0.16 – 0.67	0.001
sclmeet collapsed [Every day]	0.52	0.23 – 0.80	<0.001
stfdem	0.06	0.03 – 0.10	<0.001
stfhlth	0.08	0.04 – 0.12	<0.001
Observations	1618		
R ² / R ² adjusted	0.122 / 0.117		

Multiple regression model with a moderation

```
moderation_model <- lm(stflife ~ sclmeet_collapsed * health_collapsed + stfdem + stfhlth, data = life_sat_ned_selected)

sjPlot::tab_model(moderation_model)
```

<i>Predictors</i>	stflife		
	<i>Estimates</i>	<i>CI</i>	<i>p</i>
(Intercept)	6.66	5.85 – 7.48	<0.001
sclmeet collapsed [Several times a month]	0.64	-0.21 – 1.50	0.138
sclmeet collapsed [Once a week]	0.87	-0.02 – 1.75	0.056
sclmeet collapsed [Several times a week]	0.93	0.13 – 1.73	0.023
sclmeet collapsed [Every day]	1.01	0.18 – 1.84	0.017
health collapsed [Good]	0.35	-0.49 – 1.18	0.415
health collapsed [Fair]	-0.36	-1.25 – 0.54	0.432
health collapsed [Poor]	-1.83	-2.90 – -0.77	0.001
stfdem	0.06	0.03 – 0.10	<0.001
stfh1th	0.07	0.04 – 0.11	<0.001
sclmeet collapsed [Several times a month] × health collapsed [Good]	-0.51	-1.44 – 0.42	0.281
sclmeet collapsed [Once a week] × health collapsed [Good]	-0.97	-1.93 – -0.00	0.049
sclmeet collapsed [Several times a week] × health collapsed [Good]	-0.80	-1.67 – 0.07	0.072
sclmeet collapsed [Every day] × health collapsed [Good]	-0.78	-1.69 – 0.13	0.094
sclmeet collapsed [Several times a month] × health collapsed [Fair]	-0.51	-1.52 – 0.50	0.323
sclmeet collapsed [Once a week] × health collapsed [Fair]	-0.31	-1.37 – 0.75	0.561
sclmeet collapsed [Several times a week] × health collapsed [Fair]	-0.47	-1.41 – 0.48	0.330
sclmeet collapsed [Every day] × health collapsed [Fair]	-0.44	-1.45 – 0.56	0.387
sclmeet collapsed [Several times a month] × health collapsed [Poor]	0.23	-1.09 – 1.55	0.729
sclmeet collapsed [Once a week] × health collapsed [Poor]	-0.30	-1.67 – 1.07	0.668
sclmeet collapsed [Several times a week] × health collapsed [Poor]	0.72	-0.48 – 1.92	0.241

sclmeet collapsed [Every day] × health collapsed [Poor]

0.91 -0.50 – 2.31 0.205

Observations 1618

R² / R² adjusted 0.134 / 0.123

Comment:

- The intercept represents that the expected life satisfaction for a person with “Very good” health and “Once a month or less” meets socially is 6.66.
- People with “Good” and “Fair” health effect on life satisfaction is not statistically significant ($p > 0.05$).
- People with “Poor” health have, on average, 1.83 units lower life satisfaction than those with “Very good” health, holding other variables constant. This effect is statistically significant ($p=0.001$). As expected, “Poor” health is associated with a significant decline in life satisfaction.
- People who socially meet “Several times a month” and “Once a week” effect on life satisfaction is not statistically significant ($p > 0.05$).
- People who socially meet “Several times a week” have, on average, 0.93 units higher life satisfaction compared to those who “Once a month or less” meet socially, holding other variables constant. This effect is statistically significant ($p=0.023$).
- People who socially meet “Every day” have, on average, 1.01 units higher life satisfaction compared to those who “Once a month or less” meet socially, holding other variables constant. This effect is statistically significant ($p=0.017$). As expected, frequent social meetings are associated with higher life satisfaction.
- With each 1 unit increase in satisfaction with democracy (stfdem), life satisfaction increases by 0.06, controlling for other predictors. This effect is small but highly significant ($p < 0.001$). It suggests that people more satisfied with democracy tend to have higher life satisfaction.
- With each 1 unit increase in satisfaction with health services (stfhlth), life satisfaction increases by 0.07, controlling for other predictors. This effect is also small but highly significant ($p < 0.001$). It suggests that people more satisfied with health services tend to have higher life satisfaction.
- The interaction effect of social meetings “Several times a month” and “Good” health is not statistically significant.
- The interaction effect of social meetings “Once a week” and “Good” health is -0.97 ($p=0.049$), suggesting the effect of Good general health on life satisfaction is significantly different for people who meet socially “Once a week”. For people with “Good” health, the positive effect of meeting socially once a week on life satisfaction is 0.97 units lower than for those with “Very good” health.
- The interaction effect of social meetings “Several times a week” and “Good” health is not statistically significant.
- The interaction effect of social meetings “Every day” and “Good” health is also not statistically significant.
- Adjusted R² = 0.123 means that the model explains 12.3% of the variance in life satisfaction.
- Not all predictors in the model have a statistically significant relationship with life satisfaction; only “Poor” health, frequent social meetings (“Several times a week”, “Every day”), stfdem, stfhlth, and the interaction of “Once a week” and “Good” health are statistically significant.

Comparison of models

```
anova(additive_model, moderation_model)
```

```
## Analysis of Variance Table
##
## Model 1: stflife ~ health_collapsed + sclmeet_collapsed + stfdem + stfhlth
## Model 2: stflife ~ sclmeet_collapsed * health_collapsed + stfdem + stfhlth
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1    1608 2749.1
## 2    1596 2712.6 12    36.528 1.791 0.04462 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Comment: ANOVA shows that moderation model explains significantly more variance than additive ($F = 1.79$, $p = 0.045$), Res.Df and RSS is smaller for moderation model. It indicates that the interaction between health and social meetings improves the model fit and the relationship between health and life satisfaction depends on the level of social interaction.

```
sjPlot::tab_model(additive_model, moderation_model)
```

Predictors	stflife			stflife		
	Estimates	CI	p	Estimates	CI	p
(Intercept)	7.14	6.76 – 7.52	<0.001	6.66	5.85 – 7.48	<0.001
health collapsed [Good]	-0.38	-0.55 – -0.21	<0.001	0.35	-0.49 – 1.18	0.415

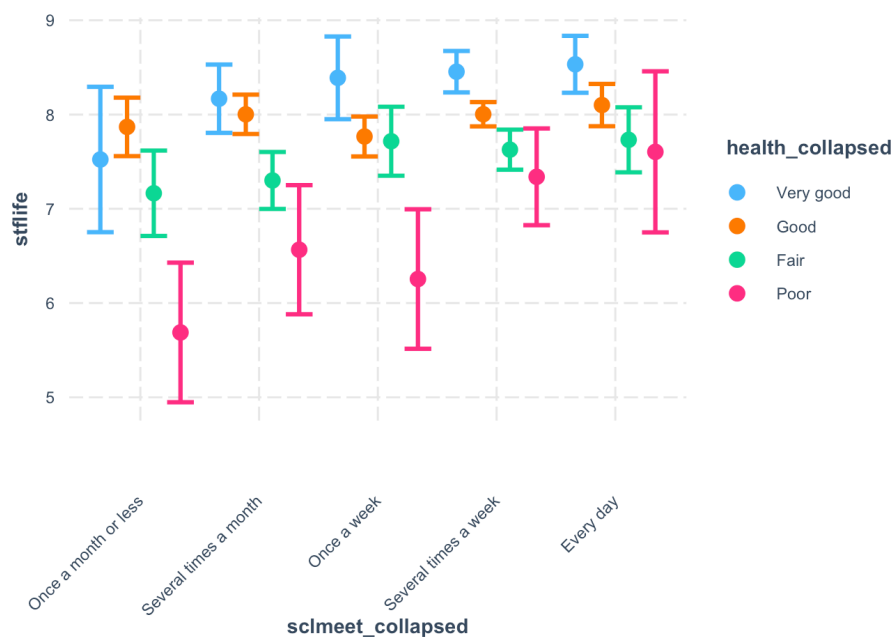
health collapsed [Fair]	-0.79	-1.00 – -0.59	<0.001	-0.36	-1.25 – 0.54	0.432
health collapsed [Poor]	-1.54	-1.88 – -1.19	<0.001	-1.83	-2.90 – -0.77	0.001
sclmeet collapsed [Several times a month]	0.25	-0.03 – 0.53	0.078	0.64	-0.21 – 1.50	0.138
sclmeet collapsed [Once a week]	0.23	-0.06 – 0.51	0.116	0.87	-0.02 – 1.75	0.056
sclmeet collapsed [Several times a week]	0.41	0.16 – 0.67	0.001	0.93	0.13 – 1.73	0.023
sclmeet collapsed [Every day]	0.52	0.23 – 0.80	<0.001	1.01	0.18 – 1.84	0.017
stfdem	0.06	0.03 – 0.10	<0.001	0.06	0.03 – 0.10	<0.001
stfhlth	0.08	0.04 – 0.12	<0.001	0.07	0.04 – 0.11	<0.001
sclmeet collapsed [Several times a month] × health collapsed [Good]				-0.51	-1.44 – 0.42	0.281
sclmeet collapsed [Once a week] × health collapsed [Good]				-0.97	-1.93 – -0.00	0.049
sclmeet collapsed [Several times a week] × health collapsed [Good]				-0.80	-1.67 – 0.07	0.072
sclmeet collapsed [Every day] × health collapsed [Good]				-0.78	-1.69 – 0.13	0.094
sclmeet collapsed [Several times a month] × health collapsed [Fair]				-0.51	-1.52 – 0.50	0.323
sclmeet collapsed [Once a week] × health collapsed [Fair]				-0.31	-1.37 – 0.75	0.561
sclmeet collapsed [Several times a week] × health collapsed [Fair]				-0.47	-1.41 – 0.48	0.330
sclmeet collapsed [Every day] × health collapsed [Fair]				-0.44	-1.45 – 0.56	0.387
sclmeet collapsed [Several times a month] × health collapsed [Poor]				0.23	-1.09 – 1.55	0.729
sclmeet collapsed [Once a week] × health collapsed [Poor]				-0.30	-1.67 – 1.07	0.668
sclmeet collapsed [Several times a week] × health collapsed [Poor]				0.72	-0.48 – 1.92	0.241
sclmeet collapsed [Every day] × health collapsed [Poor]				0.91	-0.50 – 2.31	0.205
Observations	1618			1618		
R ² / R ² adjusted	0.122 / 0.117			0.134 / 0.123		

Comment:

- The intercept represents that the expected life satisfaction for a person with “Very good” health and “Once a month or less” meets socially is 6.66 in the moderation model and 7.14 in the additive model.
- People with “Good” and “Fair” health have effects on life satisfaction that are not statistically significant ($p > 0.05$) in the moderation model, while in the additive model these effects are statistically significant and negative.
- People with “Poor” health have, on average, 1.83 units lower life satisfaction in the moderation model and 1.54 units lower in the additive model compared to those with “Very good” health, both statistically significant ($p = 0.001$ and $p < 0.001$), indicating a strong negative impact of poor health.
- Effects of social meetings “Several times a month” and “Once a week” on life satisfaction are not statistically significant in both models.
- Frequent social meetings (“Several times a week” and “Every day”) have statistically significant positive effects on life satisfaction in both models, with larger estimates in the moderation model (0.93 and 1.01) compared to the additive model (0.41 and 0.52), suggesting stronger associations when interactions are included.
- Both models show small but highly significant positive effects of satisfaction with democracy (stfdem) and health services (stfhlth) on life satisfaction, with similar estimates around 0.06–0.08.
- The moderation model includes interaction terms between social meetings and health categories; most interactions are not statistically significant except for the interaction of “Once a week” social meetings and “Good” health (estimate = -0.97, $p = 0.049$), indicating a significant moderation effect.
- Adjusted R^2 increases from 0.117 in the additive model to 0.123 in the moderation model, showing a modest improvement in explained variance by including interactions.
- Overall, the moderation model reveals nuanced moderation effects of health on the relationship between social meetings and life satisfaction that are not captured in the additive model.

Interaction plot

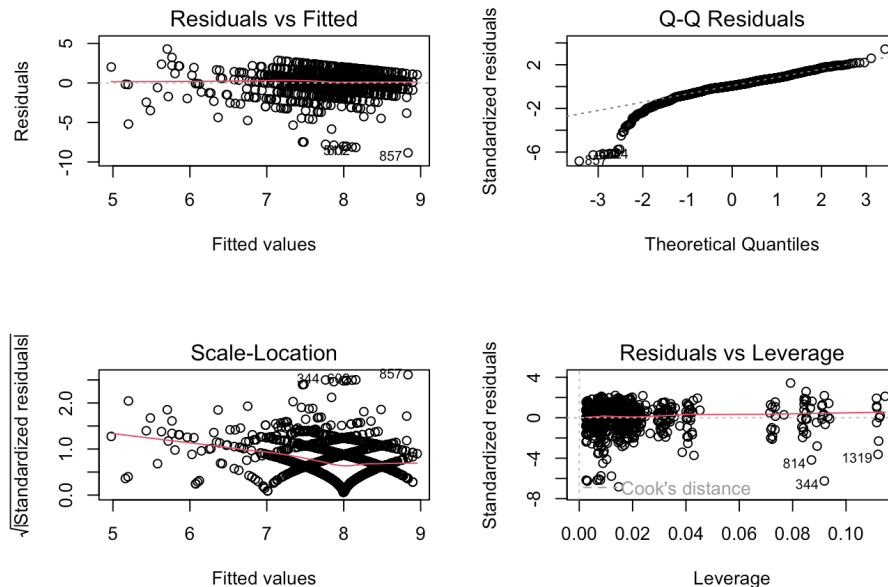
```
cat_plot(moderation_model, pred = sclmeet_collapsed, modx = health_collapsed) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    axis.title.x = element_text(vjust = -9),
    plot.margin = margin(t = 10, r = 10, b = 40, l = 10)
  )
```



Comment: Interaction plot shows that for individuals with very good or good health, life satisfaction is higher with less frequent social meetings, but decreases as the frequency of meetings increases. Conversely, for those with fair or poor health, life satisfaction is lower overall, but can slightly improve with more frequent social interactions, especially when meeting daily. This suggests that the impact of social meetings on life satisfaction depends on one's health status, with more frequent meetings potentially having a negative effect on those with better health and a positive effect on those with poorer health. We can not see clearly the interaction effect of social meetings “Once a week” and “Good” health but plot shows the tendency which confirms it.

Linear regression assumptions

```
par(mfrow = c(2,2))
plot(moderation_model)
```



```
# for homoscedasticity
bptest(moderation_model)
```

```
##
## studentized Breusch-Pagan test
##
## data: moderation_model
## BP = 40.989, df = 21, p-value = 0.005624
```

1 - In the **Residuals vs Fitted** plot, the residuals appear randomly distributed along the horizontal line at zero, which supports the assumption of linearity between predictors and the response.

2 - In the **Q-Q** plot, most residuals closely follow the diagonal line, suggesting approximate normality. However, the presence of heavy tails indicates some deviations from a perfect normal distribution.

3 - In the **Scale-Location** plot, the red line is fairly horizontal, indicating constant variance (homoscedasticity), although there is a slight decline toward the right end, which may suggest increasing or decreasing variance. To confirm, we conducted a formal test (Breusch-Pagan), and since $p < 0.05$, we reject the null hypothesis of homoscedasticity, indicating that heteroscedasticity is present.

4 - As for **Residuals vs Leverage** plot we see 3 labeled points below red line which is threshold for influential outliers, which may pull the regression.

Conclusion

Our analysis partially confirms the proposed hypotheses. As expected, better subjective health (H1) and more frequent social interactions (H2) are linked to higher life satisfaction, especially for those in poorer health and those who socialize daily or several times a week. The moderation effect (H3, H6) is only partly supported. We found a significant interaction between having "Good" health and meeting socially once a week: the positive impact of "Good" health on life satisfaction is 0.97 units lower for those who meet socially once a week compared to those with "Very good" health. This suggests that the benefit of "Good" health on life satisfaction is less pronounced among people who socialize only occasionally. One possible explanation is that individuals with "Good" (rather than "Very good") health may require more frequent social engagement to fully enjoy the benefits of their health status. Meeting socially just once a week might not be sufficient to support their well-being, thus dampening the positive effect of their health on life satisfaction.

Both trust in democratic institutions (H4) and satisfaction with the healthcare system (H5) show small but statistically significant positive effects on life satisfaction.

In simple terms, people in the Netherlands tend to feel more satisfied with their lives when they are in good health, socialize often, and trust their government and healthcare system. The influence of health on happiness may depend somewhat on how frequently people socialize, but this pattern is not consistent across all groups.

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